PROCEEDINGS A

royalsocietypublishing.org/journal/rspa

Research

Cite this article: Zrnić D, Brenn G. 2024 Weakly nonlinear shape oscillations of viscoelastic drops. *Proc. R. Soc. A* **480**: 20230887.
https://doi.org/10.1098/rspa.2023.0887

Received: 27 January 2024
Accepted: 3 July 2024

Subject Category:
Physics

Subject Areas:
fluid mechanics

Keywords:
drop-shape oscillations, weakly nonlinear analysis, viscoelastic liquid, quasi-periodic motion, onset of aperiodic motion

Author for correspondence:
Günter Brenn
e-mail: guenter.brenn@tugraz.at

Electronic supplementary material is available online at https://doi.org/10.6084/m9.figshare.c.7399609.

THE ROYAL SOCIETY PUBLISHING

Weakly nonlinear shape oscillations of viscoelastic drops

Dino Zrnić and Günter Brenn

Institute of Fluid Mechanics and Heat Transfer, Inffeldgasse 25/F, Graz 8010, Austria

GB, 0000-0001-7576-0790

Axisymmetric shape oscillations of a viscoelastic drop in a vacuum are studied by a weakly nonlinear analysis. The two-lobed initial drop deformation mode is studied. The Oldroyd-B model is used for characterizing the drop liquid rheological behaviour. The equations of motion and the solutions up to second order are developed, where elastic effects appear. Solutions of the characteristic equation are validated against the decay rate and frequency of damped shape oscillations of polymer solution drops in an acoustic levitator. The theory shows enhancing or dampening of the nonlinear behaviour and enhanced mode coupling, as compared with the Newtonian case. The study reveals an excess time in the prolate shape and a frequency change, together with a quasi-periodicity of the oscillations. The Fourier power spectra of traces of the drop north pole position in time show mode coupling as a nonlinear effect. At moderate stress-relaxation Deborah number, the resultant drop motion for large Ohnesorge number, suggesting aperiodic drop behaviour, may nonetheless be oscillatory, where the drop elasticity is owing to the liquid bulk elasticity instead of surface tension. A method is developed to predict the oscillatory or aperiodic behaviour of a drop of a given size from a rheologically characterized viscoelastic Oldroyd-B liquid.

1. Introduction

For more than one and a half centuries, shape oscillations of drops have been scientifically investigated. The motivations are fundamental interest and the influence on mass, momentum and energy transport across the drop surface. Rayleigh presented in the appendix to his paper on the capillary phenomena of jets an analysis of linear shape oscillations of an inviscid drop in a vacuum around a spherical equilibrium state [1].

Rayleigh obtained an equation for the angular oscillation frequency of the drop deformed for a given deformation mode m assuming natural values greater than unity corresponding to the number of lobes on the drop surface. A generalization for viscous drop liquid and an ambient medium characterized by its density is owing to Lamb [2,3]. The analysis predicts the Ohnesorge number $Oh_0 = \mu/(\sigma a \rho)^{1/2}$ of the drop (dynamic viscosity μ, surface tension σ, drop radius a and density ρ) at the onset of aperiodic motion. Chandrasekhar analysed small-amplitude shape oscillations of a viscous, self-gravitating globe in a vacuum, developing the characteristic equation of the globe to determine the complex angular oscillation frequency [4]. Reid showed that the results of this analysis are equivalent for the oscillations of a drop if the restoring gravity force is replaced by the force owing to surface tension [5]. Another generalizing step in the analysis is owing to Miller & Scriven [6] and Basaran [7], who accounted for both the viscous and the inertial influences from the ambient medium hosting the viscous oscillating drop. In the work [6], the interfacial dilatational viscosity and elasticity of the drop were addressed for the first time. The important aspect of the initiation of the oscillations was studied by Prosperetti [8]. He analysed the drop-shape oscillations, formulating an initial-value problem. The solutions show that the normal-mode approach may predict aperiodic drop motion in a range of Ohnesorge numbers, where the solutions of the initial-value problem show the motion to turn into periodic in time [8]. The most important results from analyses of drop-shape oscillations are the angular frequency and decay rate of the oscillations, together with the time-dependent shapes of the drops.

One of the first theoretical analyses of small-amplitude axisymmetric shape oscillations of a viscoelastic liquid drop in microgravity is owing to Khismatullin & Nadim [9]. The authors derived the characteristic equation for a viscoelastic drop and found that the equation has an infinite number of roots, depending critically on the values of the stress relaxation and deformation retardation times, as well as the surface tension. Asymptotic analyses of the characteristic equation in the low- and high-viscosity limits and for low, moderate and large elasticities are performed, showing the periodic and damped or aperiodic behaviours of the drops. A more recent study numerically analysed freely decaying shape oscillations of viscoelastic liquid drops in another viscoelastic host liquid, using a Navier–Stokes/Oldroyd-B immersed boundary technique [10]. For low viscosity, the fluid viscoelasticity was found to modulate the drop oscillatory shape relaxation, raising the oscillation frequency and reducing the decay rate, when the fluid relaxation time is above a critical value. The oscillation behaviour of small drops of a viscoelastic, shear-thinning aqueous solution applied in drop-on-demand inkjet printing was experimentally investigated by Hoath *et al*. [11]. The effects of surface charge and the electrical properties of the viscoelastic liquids on drop oscillation and instability characteristics were investigated by Li *et al*. [12].

Properties of drop-shape oscillations and their dependency on the physical properties of the drop liquid may be used for liquid material characterization. Using this oscillating drop method, the liquid surface tension against the ambient gas [13–15] and the interfacial tension between immiscible liquids [16], as well as the liquid dynamic viscosity [17–19], are measured. For drops from viscoelastic liquids, including surfactant solutions, surface rheological properties were measured [20–24]. The shape oscillations of a drop in the presence of surfactants were studied by Lu & Apfel [25]. The authors showed that, with interfacial Gibbs elasticity and viscosity, shape oscillations are dampened more strongly than with just an interfacial tension as the interfacial property. Investigations of the dynamic surface tension of a drop against its ambient gaseous host medium, using a growing-drop technique, are owing to Zhang *et al*. [26]. Shape oscillations of the pendant drop prevent the measurements in the first 20 ms after the formation of the interface. Kovalchuk *et al*. [27] provided a comprehensive review of oscillating drop and bubble techniques as published before the year 2001. The paper Ravera *et al*. [28] updated this with a focus on interfacial dilational rheology by the oscillating bubble and drop methods. The conclusion was that the accuracy achieved with these methods allows for a reliable measurement of the dilational viscoelasticity of interfaces as a function of the frequency. For Newtonian drops, the dynamic surface tension, and for shear-thinning drops, both the transient shear viscosity and the dynamic surface tension were measured, using the oscillating drop method in Yang *et al*. [29]. Measurements were performed on free-falling, oscillating drops. The authors also showed that the dynamic viscosity greatly affects the liquid behaviour at high shear.

The study by Brenn & Plohl [30] used the damped drop-shape oscillation method for measuring the deformation retardation time of polymeric liquids, thus introducing a drop-oscillation-based experimental method for determining this polymeric timescale. In their studies, they relied on Khismatullin & Nadim [9] and Brenn & Teichtmeister [31] in theoretically analysing small-amplitude axisymmetric shape oscillations of viscoelastic drops in a gas using the Jeffreys model as the rheological constitutive equation (RCE) of the liquid. The study showed that the measured values of the deformation retardation time corresponding to the damped shape oscillations investigated deviate strongly from the values predicted by the viscoelastic stress splitting method, which is often used in simulations of viscoelastic liquid flow. Ponce-Torres *et al*. [32] studied the break-up of surfactant-laden drops, concluding from shape oscillations of satellite droplets on their surfactant content. Drop-shape oscillations induced by acoustic drop levitation were used for measuring the rheological material behaviour of blood in the work [33]. The technique makes it possible to assess blood viscosity, including its changes, and can be used for monitoring blood rheology in sickle cell and other haematological diseases. Most recently, the frequency-response diagram of a soft viscoelastic drop driven to shape oscillations was proposed as the basis for a rheometric experiment, using either an ultrasonically levitated drop or to be conducted in a microgravity environment [34]. The drop rheological behaviour was characterized using the Kelvin–Voigt and Maxwell models for the soft gels and polymeric fluids, respectively. The frequency-response diagram from the theoretical analysis is the basis for the use of the drop-oscillation experimental set-up as a 'drop vibration rheometer', which is proposed for future experiments. These findings show that the literature on nonlinear shape oscillations for viscoelastic drops is sparse. For further understanding of the measured nonlinear drop-shape oscillations described above, a reliable theoretical method is needed to describe the time evolution of the drop shapes for viscoelastic liquids.

In the present work, we analyse in detail the time behaviour of nonlinear shape oscillations of viscoelastic drops. Steered by our earlier work [35,36], we study nonlinear effects in shape oscillations of viscoelastic drops in a vacuum, which are initially deformed by the two-lobed mode $m = 2$. The basis is a weakly nonlinear analysis of the oscillations, which we restrict to axisymmetric drop shape. The analysis is carried to the second order of approximation since elastic effects appear at that order. This kind of analysis was not done in the literature before. The viscoelasticity of the drop liquid is characterized by the Oldroyd-B RCE. The analysis is based on series expansions of the flow field variables and the drop shape with respect to a deformation parameter. The nonlinear phenomena, which consist of an asymmetry of the times spent in different states of drop deformation and an oscillation frequency change for varying initial drop deformation, are revealed, covering a wide range of drop Ohnesorge numbers as found in many technical applications. The frequencies of the oscillatory motions are determined by the characteristic equation of the drop. The quasi-periodicity of the motion, which results from mode coupling, is unveiled. The time dependency of the solutions for the flow field variables and the drop shape, at a given approximation order, are governed by frequencies in products of solutions of the respective lower-order approximations. The results are relevant for mass, momentum and energy transport across the oscillating drop surface. Applications in production technologies, such as inkjet printing [37], containerless materials processing based on single drops [38], spray cooling [39,40], in-air microfluidics [41,42] and food production, will benefit from the results, since they allow for a more accurate prediction of evaporation rates, drag coefficients and heating or cooling rates of the drops.

In the following section, the problem is formulated, and the equations of motion with their boundary and initial conditions are derived up to the second order of approximation. In §3, the solutions of the governing equations for the respective orders of approximation are presented. Results from the analysis are presented and discussed in §4. In §5, the conclusions from the work are summarized.

2. Formulation of the problem

We study the weakly nonlinear shape oscillations of a viscoelastic liquid drop as sketched in figure 1. The drop is axisymmetric with respect to the azimuthal coordinate ϕ of the spherical coordinate system. The liquid is treated as incompressible, and its viscoelastic rheological behaviour is represented by the Oldroyd-B model. The dynamic influence from an ambient medium is neglected, i.e. we analyse the drop motion in a vacuum. Body forces are not accounted for, since in many processes with droplets the Froude number $Fr = (\sigma/\rho g a^2)^{1/2}$ is large enough to allow for this neglect. The problem is formulated in spherical coordinates (r, θ, ϕ) to account for its geometry.

The equations of motion with their initial and boundary conditions are non-dimensionalized with the undeformed drop radius a, the capillary time scale $(\rho a^3/\sigma)^{1/2}$, the capillary pressure σ/a and the extra stress $\mu_0(\sigma/\rho a^3)^{1/2}$ for length, time, pressure and extra stress, respectively. Here, ρ is the liquid density, σ the air–liquid interfacial tension and μ_0 the zero-shear viscosity of the drop liquid. The drop surface is described as the place where $r_s(\theta, t) = 1 + \eta(\theta, t)$ (cf. figure 1), with the non-dimensional deformation η against the undisturbed spherical shape.

For the problem at hand, the equation of continuity and the momentum balances in the radial and polar angular directions, r and θ, read

$$\frac{1}{r^2}\frac{\partial}{\partial r}(r^2 u_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(u_\theta \sin\theta) = 0, \tag{2.1}$$

$$\begin{aligned}\frac{\partial u_r}{\partial t} + u_r\frac{\partial u_r}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_r}{\partial\theta} - \frac{u_\theta^2}{r} = &-\frac{\partial p}{\partial r} + \\ &+ Oh_0\left[\frac{1}{r^2}\frac{\partial}{\partial r}(r^2\tau_{rr}) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\tau_{r\theta}\sin\theta) - \frac{\tau_{\theta\theta}+\tau_{\phi\phi}}{r}\right],\end{aligned} \tag{2.2}$$

$$\begin{aligned}\frac{\partial u_\theta}{\partial t} + u_r\frac{\partial u_\theta}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_\theta}{\partial\theta} + \frac{u_r u_\theta}{r} = &-\frac{1}{r}\frac{\partial p}{\partial\theta} + \\ &+ Oh_0\left[\frac{1}{r^2}\frac{\partial}{\partial r}(r^2\tau_{r\theta}) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\tau_{\theta\theta}\sin\theta) + \frac{\tau_{r\theta}}{r} - \frac{\cot\theta}{r}\tau_{\phi\phi}\right],\end{aligned} \tag{2.3}$$

where $Oh_0 = \mu_0/(\sigma a\rho)^{1/2}$ is the Ohnesorge number, the characteristic dimensionless parameter measuring viscous against capillary time scales. The viscoelastic liquid material is characterized by the non-dimensional RCE of the Oldroyd-B fluid

$$\boldsymbol{\tau} + De_1\overset{\triangledown}{\boldsymbol{\tau}} = 2\left(\boldsymbol{D} + De_2\overset{\triangledown}{\boldsymbol{D}}\right), \tag{2.4}$$

where $\boldsymbol{\tau}$ and $\boldsymbol{D}$ are the extra-stress and rate-of-deformation tensors, respectively, the latter defined with the velocity gradient tensor $\vec{\nabla}\vec{v}$ as

$$\boldsymbol{D} = \frac{1}{2}\left(\vec{\nabla}\vec{v} + \vec{\nabla}\vec{v}^{\,T}\right). \tag{2.5}$$

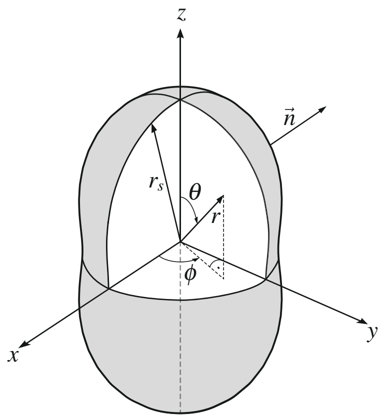

Figure 1. Sketch of the geometry of a liquid drop under deformation at mode 2 (adapted from Smuda *et al.* [43]).

The Deborah numbers $De_1 = \lambda_1(\sigma/\rho a^3)^{1/2}$ and $De_2 = \lambda_2(\sigma/\rho a^3)^{1/2}$ represent the non-dimensional stress relaxation and deformation retardation times of the liquid, respectively. The upper-convected derivative $\overset{\triangledown}{\boldsymbol{A}}$ of a tensor $\boldsymbol{A}$ is given as

$$\overset{\triangledown}{\boldsymbol{A}} = \frac{d\boldsymbol{A}}{dt} - \vec{\nabla}\vec{v} \cdot \boldsymbol{A} - \boldsymbol{A} \cdot \vec{\nabla}\vec{v}^{\,T}, \tag{2.6}$$

where $d\boldsymbol{A}/dt$ is the material derivative. The alternative of modelling the drop liquid as a second-order fluid would represent the time behaviour of the extra stress differently.

The equations of change are solved by applying initial conditions and boundary conditions. The material rate of deformation of the drop surface is equal to the radial velocity component of the deformed drop surface. The kinematic boundary condition therefore reads

$$u_r = \frac{d\eta}{dt} = \frac{\partial \eta}{\partial t} + \frac{u_\theta}{r}\frac{\partial \eta}{\partial \theta} \quad \text{at } r = 1 + \eta. \tag{2.7}$$

The shear stress at the surface of a drop in a vacuum vanishes since momentum transfer across the drop surface is not possible. The related dynamic boundary condition therefore reads

$$\left(\vec{n} \cdot \boldsymbol{\tau}\right) \times \vec{n} = \vec{0} \quad \text{at } r = 1 + \eta. \tag{2.8}$$

The unit normal vector $\vec{n}$ on the drop surface, pointing outward, is determined by

$$\vec{n} = \frac{1}{|\vec{\nabla}F|}\vec{\nabla}F \quad \text{with } F = r - 1 - \eta(t, \theta) = 0. \tag{2.9}$$

The tensor of deformation-induced stress in equation (2.8) is formulated for an incompressible Oldroyd-B fluid. Furthermore, the normal stress on the drop surface, which consists of pressure, viscoelastic normal stress and capillary stress, vanishes for the same reason as shear stress. The normal-stress boundary condition therefore is obtained as

$$-p + Oh_0\left(\vec{n} \cdot \boldsymbol{\tau}\right) \cdot \vec{n} + \left(\vec{\nabla} \cdot \vec{n}\right) = 0 \quad \text{at } r = 1 + \eta. \tag{2.10}$$

For the deformed drop, the divergence of the normal unit vector on the drop surface in this equation is obtained as [35]

$$\left(\vec{\nabla}\cdot\vec{n}\right)=\frac{1}{r}\frac{2+3\left(\frac{1}{r}\frac{\partial\eta}{\partial\theta}\right)^2}{\left[1+\left(\frac{1}{r}\frac{\partial\eta}{\partial\theta}\right)^2\right]^{3/2}}-\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left[\frac{\frac{\partial\eta}{\partial\theta}}{\left(1+\left(\frac{1}{r}\frac{\partial\eta}{\partial\theta}\right)^2\right)^{1/2}}\sin\theta\right]\quad\text{at}\quad r=1+\eta. \tag{2.11}$$

In a weakly nonlinear analysis of the oscillating drop problem, the flow field variables and the drop surface shape are formulated as series expansions with respect to a small deformation parameter, denoted η_0. For details, the reader is referred to our earlier work [36]. The formulation, e.g. for the extra stress tensor, reads

$$\boldsymbol{\tau}=\boldsymbol{\tau}_1\eta_0+\boldsymbol{\tau}_2\eta_0^2+\dots\,. \tag{2.12}$$

The series expansions converge for sufficiently small deformation parameter η_0. In the weakly nonlinear analysis, the boundary conditions are formulated on the deformed drop surface. The values of the field variables on the deformed drop surface are obtained from Taylor expansions, such as, for example, for $\boldsymbol{\tau}$

$$\boldsymbol{\tau}\big|_{r=1+\eta}=\boldsymbol{\tau}\big|_{r=1}+\left.\frac{\partial\boldsymbol{\tau}}{\partial r}\right|_{r=1}\eta+\dots\,. \tag{2.13}$$

Applying this framework to the equations of motion, the RCE and the boundary conditions and representing the flow properties and their derivatives as given for the example of $\boldsymbol{\tau}$ in (2.13), sets of first- and second-order equations of motion and boundary conditions are obtained, which consist of terms with the deformation parameter η_0 to the first and second powers, respectively. All the terms together, which are linear in the parameter, represent the first-order equations, and the terms with quadratic η_0 are the second-order equations.

The analysis assumes that the non-equilibrium starting the drop motion consists of an initial drop surface that differs from the spherical shape. This drop shape at the beginning of the motion is described by a Legendre polynomial of degree m with the amplitude η_0. Furthermore, it is assumed that the surface is initially at rest. The mode number m equals the number of lobes of the drop surface along the polar angle θ. When calculating the volume of the deformed drop, the deformed drop shape is obtained as

$$\begin{aligned}r_s(\theta,0)&=1+\eta(\theta,0)=\left(1+\frac{3\eta_0^2}{2m+1}+\frac{1}{2}\eta_0^3\int_{-1}^{1}P_m(\cos\theta)^3d(\cos\theta)\right)^{-1/3}+\eta_0P_m(\cos\theta)\\&\approx 1+\eta_0P_m(\cos\theta)-\eta_0^2\frac{1}{2m+1}-\frac{\eta_0^3}{6}\int_{-1}^{1}P_m(\cos\theta)^3d(\cos\theta)\mp\dots\,.\end{aligned} \tag{2.14}$$

Each term with η_0 to a given power represents the factor ensuring volume conservation at the given approximation order. In the solutions for the drop-shape oscillations, the angular frequencies for the oscillation modes zero and unity are needed, which are not readily found in the characteristic equation of the drop. This problem can be solved using the fact that the centre of drop mass does not move away from its initial position

$$z_m=\frac{3}{8}\int_{-1}^{1}\left[1+4\eta_0\eta_1+\eta_0^2\left(6\eta_1^2+4\eta_2\right)+\eta_0^3\left(4\eta_1^3+12\eta_1\eta_2+4\eta_3\right)+\dots\right]x\,dx\,, \tag{2.15}$$

where $x=\cos\theta$. The physical reason is that there is no resultant force acting on the drop that could cause such a motion. This will be detailed more in the course of the analysis below.

(a) First-order equations

The equations for the two different approximation orders are presented here without details of their derivation, in order not to repeat details given in previous publications [35,36]. The first-order equations of change read

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 u_{r1}\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(u_{\theta 1}\sin\theta) = 0, \tag{2.16}$$

$$\begin{aligned}\frac{\partial u_{r1}}{\partial t} &= -\frac{\partial p_1}{\partial r} + \\ &+ Oh_0\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\tau_{rr1}\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\tau_{r\theta 1}\sin\theta) - \frac{\tau_{\theta\theta 1}+\tau_{\phi\phi 1}}{r}\right],\end{aligned} \tag{2.17}$$

$$\begin{aligned}\frac{\partial u_{\theta 1}}{\partial t} &= -\frac{1}{r}\frac{\partial p_1}{\partial\theta} + \\ &+ Oh_0\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\tau_{r\theta 1}\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\tau_{\theta\theta 1}\sin\theta) + \frac{\tau_{r\theta 1}}{r} - \frac{\cot\theta}{r}\tau_{\phi\phi 1}\right].\end{aligned} \tag{2.18}$$

The RCE of the first order becomes

$$\boldsymbol{\tau}_1 + De_1\frac{\partial\boldsymbol{\tau}_1}{\partial t} = 2\left(\boldsymbol{D}_1 + De_2\frac{\partial\boldsymbol{D}_1}{\partial t}\right). \tag{2.19}$$

The first-order boundary conditions at $r = 1$ read

$$u_{r1} = \frac{\partial\eta_1}{\partial t}, \qquad \tau_{r\theta 1} = 0, \tag{2.20}$$

$$-p_1 + Oh_0\tau_{rr1} - \left(2\eta_1 + \frac{\partial\eta_1}{\partial\theta}\cot\theta + \frac{\partial^2\eta_1}{\partial\theta^2}\right) = 0, \tag{2.21}$$

and the first-order initial conditions are

$$\eta_1(\theta, 0) = P_m(\cos\,\theta), \qquad \frac{\partial\eta_1}{\partial t}(\theta, 0) = 0. \tag{2.22}$$

(b) Second-order equations

The second-order equations of change read

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 u_{r2}\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(u_{\theta 2}\sin\theta) = 0, \tag{2.23}$$

$$\begin{aligned}&\frac{\partial u_{r2}}{\partial t} - Oh_0\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\tau_{rr2}\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\tau_{r\theta 2}\sin\theta) - \frac{\tau_{\theta\theta 2}+\tau_{\phi\phi 2}}{r}\right] + \frac{\partial p_2}{\partial r} = \\ &\quad \frac{u_{\theta 1}^2}{r} - u_{r1}\frac{\partial u_{r1}}{\partial r} - \frac{u_{\theta 1}}{r}\frac{\partial u_{r1}}{\partial\theta},\end{aligned} \tag{2.24}$$

$$\begin{aligned}&\frac{\partial u_{\theta 2}}{\partial t} - Oh_0\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\tau_{r\theta 2}\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\tau_{\theta\theta 2}\sin\theta) + \frac{\tau_{r\theta 2}}{r} - \frac{\cot\theta}{r}\tau_{\phi\phi 2}\right] + \frac{1}{r}\frac{\partial p_2}{\partial\theta} = \\ &\quad - u_{r1}\frac{\partial u_{\theta 1}}{\partial r} - \frac{u_{\theta 1}}{r}\frac{\partial u_{\theta 1}}{\partial\theta} - \frac{u_{r1}u_{\theta 1}}{r}.\end{aligned} \tag{2.25}$$

The RCE of the second order is

$$\boldsymbol{\tau}_2 + De_1\frac{\partial \boldsymbol{\tau}_2}{\partial t} - 2\left(\boldsymbol{D}_2 + De_2\frac{\partial \boldsymbol{D}_2}{\partial t}\right) = \\ - De_1\left[\left(\vec{v}_1\cdot\vec{\nabla}\right)\boldsymbol{\tau}_1 - \vec{\nabla}\vec{v}_1\cdot\boldsymbol{\tau}_1 - \boldsymbol{\tau}_1\cdot\vec{\nabla}\vec{v}_1^{\,T}\right] + \\ + 2De_2\left[\left(\vec{v}_1\cdot\vec{\nabla}\right)\boldsymbol{D}_1 - \vec{\nabla}\vec{v}_1\cdot\boldsymbol{D}_1 - \boldsymbol{D}_1\cdot\vec{\nabla}\vec{v}_1^{\,T}\right]. \tag{2.26}$$

The second-order boundary conditions at $r = 1$ are

$$u_{r2} - \frac{\partial \eta_2}{\partial t} = \frac{u_{\theta 1}}{r}\frac{\partial \eta_1}{\partial \theta} - \eta_1\frac{\partial u_{r1}}{\partial r}, \tag{2.27}$$

$$\tau_{r\theta 2} = -\eta_1\frac{\partial \tau_{r\theta 1}}{\partial r} - (\tau_{rr1} - \tau_{\theta\theta 1})\frac{1}{r}\frac{\partial \eta_1}{\partial \theta}, \tag{2.28}$$

$$-p_2 + Oh_0\tau_{rr2} - \left(2\eta_2 + \frac{\partial \eta_2}{\partial \theta}\cot\theta + \frac{\partial^2 \eta_2}{\partial \theta^2}\right) = \eta_1\frac{\partial p_1}{\partial r} \\ - Oh_0\left[\eta_1\frac{\partial \tau_{rr1}}{\partial r} - 2\frac{1}{r}\frac{\partial \eta_1}{\partial \theta}\tau_{r\theta 1}\right] - \left(2\eta_1^2 + 2\eta_1\frac{\partial \eta_1}{\partial \theta}\cot\theta + 2\eta_1\frac{\partial^2 \eta_1}{\partial \theta^2}\right) \tag{2.29}$$

and the initial conditions of the second order are

$$\eta_2(\theta, 0) = -\frac{1}{2m+1}, \quad \frac{\partial \eta_2}{\partial t}(\theta, 0) = 0. \tag{2.30}$$

In the next step, the solutions of the first-order equations will be presented.

3. Solutions of the governing equations

(a) First-order solutions

The linear problem is governed by the first-order equations. For viscous, Newtonian drop liquid, well-known linear solutions are owing to Lamb [2,3] and Chandrasekhar [4]. The two-dimensional flow field allows the method of the Stokesian stream function to be applied to determine the velocity and pressure fields. The stream function $\psi(r, \theta, t)$ is related to the two velocity components u_{r1} and $u_{\theta 1}$ by [44]

$$u_{r1} = -\frac{1}{r^2\sin\theta}\frac{\partial \psi}{\partial \theta} \quad \text{and} \quad u_{\theta 1} = \frac{1}{r\sin\theta}\frac{\partial \psi}{\partial r}. \tag{3.1}$$

These formulations ensure that the first-order velocity field satisfies the continuity equation.

The first-order drop surface deformation is governed by the Legendre polynomial of the initial deformation. The solution is therefore sought in the form

$$\eta_1(\theta, t) = \hat{\eta}_1 P_m(\cos\theta)e^{-\alpha_m t}, \tag{3.2}$$

with the first-order initial surface amplitude $\hat{\eta}_1$ and the complex angular frequency α_m for the deformation mode m.

To find the first-order solutions of the equations of motion, we first solve the RCE of the first order. Given the exponential dependency of all the flow field variables on time via $\exp(-\alpha_m t)$, we find for the extra-stress tensor of first order

$$\boldsymbol{\tau}_1 = 2\frac{1 - De_2\alpha_m}{1 - De_1\alpha_m}\boldsymbol{D}_1 =: 2\beta_1\boldsymbol{D}_1. \tag{3.3}$$

This means that, at first order, the extra-stress tensor of the viscoelastic fluid differs from the Newtonian material just by a frequency-dependent factor β_1 in front of the rate-of-deformation tensor. This fluid is therefore formally identical to a Newtonian one, so that all the first-order results obtained in our previous paper [36] apply here, just with the Ohnesorge number Oh_0 in those equations replaced by the modified Ohnesorge number $Oh_v =: \beta_1 Oh_0$. We therefore apply those previous results as follows.

The curl of the vectorial momentum equation consisting of equations (2.17) and (2.18) yields the partial differential equation

$$\left(\frac{1}{Oh_v}\frac{\partial}{\partial t} - E_s^2\right)\left(E_s^2\psi\right) = 0, \tag{3.4}$$

for the stream function, where the operator is given as [44]

$$E_s^2 = \frac{\partial^2}{\partial r^2} + \frac{\sin\theta}{r^2}\frac{\partial}{\partial\theta}\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\right). \tag{3.5}$$

The solution is the stream function $\psi = \psi_1 + \psi_2$ [45], where

$$\psi_1 = C_{1m} r^{m+1} \sin^2\theta P_m'(\cos\theta) e^{-\alpha_m t} \quad \text{and} \quad \psi_2 = C_{2m} q r j_m(qr) \sin^2\theta P_m'(\cos\theta) e^{-\alpha_m t}. \tag{3.6}$$

In these solutions, $P_m'(\cos\theta)$ is the first derivative of the Legendre polynomial P_m with respect to its argument. The symbol j_m denotes a spherical Bessel function of the first kind and order m. The parameter q in its argument equals $\sqrt{\alpha_m/Oh_v}$. The velocity components in the radial and the polar angular directions are obtained as

$$u_{r1} = -\left[C_{1m} r^{m-1} + C_{2m} q^2 \frac{j_m(qr)}{qr}\right] m(m+1) P_m(\cos\theta) e^{-\alpha_m t} \tag{3.7}$$

and

$$u_{\theta 1} = \left[C_{1m}(m+1) r^{m-1} + C_{2m} q^2 \left((m+1)\frac{j_m(qr)}{qr} - j_{m+1}(qr)\right)\right] \sin\theta P_m'(\cos\theta) e^{-\alpha_m t}, \tag{3.8}$$

respectively. The two integration constants C_{1m} and C_{2m} are determined by the first-order kinematic and zero-shear stress boundary conditions and read

$$C_{1m} = \frac{\hat{\eta}_1 \alpha_m}{m(m+1)}\left[1 + \frac{2\left(m^2-1\right)}{2q j_{m+1}(q)/j_m(q) - q^2}\right] \qquad C_{2m} = -\frac{2(m-1)\hat{\eta}_1\alpha_m}{mq[2q j_{m+1}(q) - q^2 j_m(q)]}. \tag{3.9}$$

The first-order pressure field is obtained by integration of the momentum equations as

$$p_1 = -C_{1m}(m+1)\alpha_m r^m P_m(\cos\theta) e^{-\alpha_m t}. \tag{3.10}$$

The solutions for the velocity and pressure fields allow the boundary condition (2.21) for the normal stress to be formulated, yielding the characteristic equation of the drop

$$\frac{\alpha_{m,0}^2}{\alpha_m^2} = \frac{2(m^2-1)}{q^2 - 2q j_{m+1}/j_m} - 1 + \frac{2m(m-1)}{q^2}\left[1 + \frac{2(m+1) j_{m+1}/j_m}{2 j_{m+1}/j_m - q}\right]. \tag{3.11}$$

This equation determines the complex angular oscillation frequency α_m, where we have denoted the non-dimensional Rayleigh frequency $\alpha_{m,0} = [m(m-1)(m+2)]^{1/2}$. The equation is formally identical to the results of Lamb, Chandrasekhar [2,4] and Khismatullin & Nadim [9]. The spherical Bessel functions are taken at the value q of their arguments. We can express the square of the argument, q^2, as

$$q^2 = \alpha_{m,0}\frac{\Omega_m}{Oh_0}\frac{1 - \Omega_m De_1 \alpha_{m,0}}{1 - \Omega_m De_2 \alpha_{m,0}}, \tag{3.12}$$

Table 1. Measured properties of the aqueous Praestol 2500 solutions and measured drop radius. Density $\rho = 10^3 kg/m^3$.

SMF (wt%)	μ_0 (Pa s)	σ (N/m)	λ_1 (s)	a (m)	Oh_0 [1]
0.3	0.043	0.073	0.078	0.925 x 10−3	0.16
0.5	0.35	0.074	0.14	1.08 x 10−3	1.238

where $\Omega_m = \alpha_m/\alpha_{m,0}$. This form is identical to the results by Brenn & Teichtmeister [31] and by Khismatullin & Nadim [9]. For $De_1 = 0$ and $De_2 = 0$, equation (3.12) reduces to the Newtonian case.

In the following section, the solutions of the characteristic equation will be shown and their importance will be addressed. A method of finding the appropriate solutions with the use of experiments will be proposed.

(b) Solutions of the characteristic equation and first-order flow fields

For identifying solutions of the characteristic equation representing correctly the time behaviour of oscillating drops, both in terms of frequency and decay rate, shape oscillations of individual drops from aqueous solutions of the polyacrylamide Praestol 2500, levitated in an ultrasonic resonator, were investigated experimentally [30]. The measured zero-shear viscosity μ_0, surface tension σ and stress-relaxation time λ_1, characterizing the drop liquid for different solute mass fractions (SMF), together with the drop radius a, are given in table 1. Owing to the small solute contents, the density ρ for all SMF equals 1000 kg/m3. These data are used to compute the Deborah numbers De_1. The deformation retardation time λ_2 will be computed from the Deborah number De_2 selected for solving the characteristic equation of the drop. The procedure will be explained in detail below.

For the given drop radii and liquids with SMF of 0.3 and 0.5 wt%, the values of the Deborah number De_1 are 23.71 and 33.93, respectively. The Ohnesorge numbers for the former and latter cases are $Oh_0 = 0.16$ and $Oh_0 = 1.238$, respectively. The solutions of the characteristic equation are non-dimensional complex angular frequencies Ω_m, where m is the deformation mode. The imaginary and real parts of the complex angular frequency are the frequency and the decay rate of the drop-shape oscillations, respectively. They are shown in figure 2 as functions of Oh_0 for the deformation modes $m = 2$ and $m = 4$. In order to show the behaviour of the drops up to aperiodic states, the Ohnesorge number ranges extend to 8 and 30 in figure 2*a,b,c,d*, respectively.

For fixed Deborah and Ohnesorge numbers, the characteristic equation yields an infinite number of roots [9]. We show only one pair of solutions since the others assume large real values, which are not relevant since the corresponding motions disappear rapidly. The remaining problem is to select a pair of complex conjugate values appropriate for representing the drop-shape oscillations observed in the experiments. For doing this, De_2 is varied such that the decay rate of the oscillations agrees with the value from the experiment. The procedure of selection will be explained in the next paragraph. Figure 3 shows traces of the drop north pole obtained from the experiments with the two viscoelastic liquid drops in table 1. For comparison, in the same figure, we show the first-order solutions of the drop north pole position $r_s(0, t)$ as a function of time. Frequency and decay rate of the linear solutions are obtained from the complex conjugate angular frequency for the corresponding liquid shown in figure 2. The surface deformation amplitudes and phase angles at are set according to the experimental data. Length and time in the experimental data are non-dimensionalized with the droplet radius and capillary timescale, respectively. The theoretical and experimental data agree very well within the uncertainty of the measured drop radius of, which is shown by the error bars.

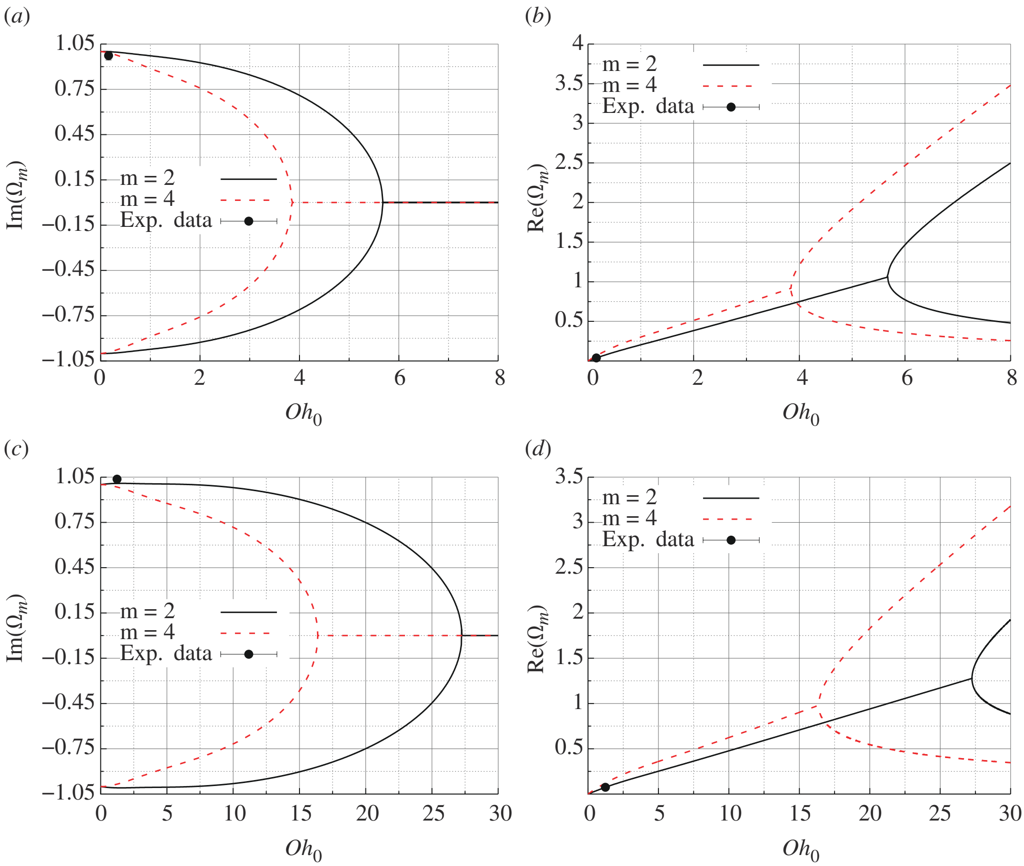

Figure 2. Solutions of the characteristic equation for polymer mass factions of (*a*,*b*) 0.3 wt% ($De_1 = 23.71$) and (*c*,*d*) 0.5 wt% ($De_1 = 33.93$), as functions of the Ohnesorge number. (*a*,*c*) Imaginary and (*b*,*d*) real parts of the non-dimensional frequency Ω_m.

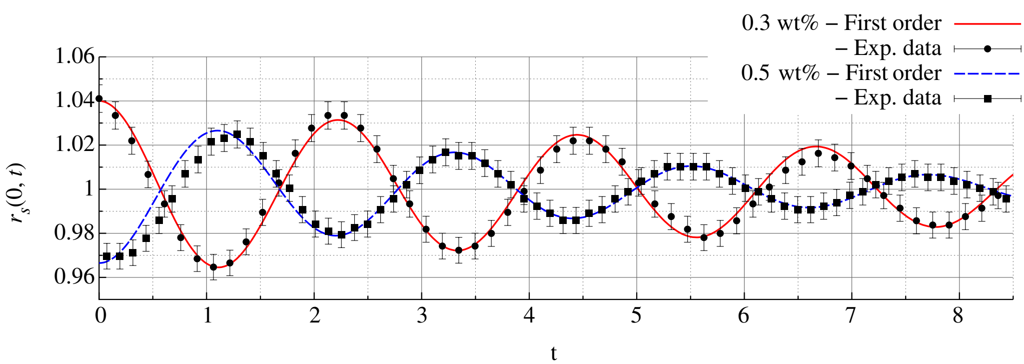

Figure 3. The trace of the drop north pole in time obtained from theory for $m = 2$ and experimental data.

In figure 2, we show for each mode m only the pair of complex conjugate values for the given range of Ohnesorge numbers. For calculation of the presented solutions, the value of the Deborah number De_2 is selected such that, for the fixed Ohnesorge number Oh_0 and Deborah number De_1, the real part of the solution Ω_2 matches the experimental non-dimensional decay rate for the deformation mode $m = 2$. This applies the oscillating drop method of Brenn & Plohl [30] in a simplified manner, laying the focus on the decay rate. The selected Deborah numbers are $De_2 = 3.5$ and $De_2 = 1.26$ for the two drops in table 1 with SMF of 0.3 and 0.5 wt%, respectively. The former and latter numbers, with the measured properties in table 1, yield the deformation retardation times λ_2 of 0.0115 and 0.0052 s, respectively. For the same drops, the measured oscillation frequencies are $f = 133.28$ Hz and $f = 113.12$ Hz, respectively, and the decay rates are $\alpha_{2,r} = 33.13$ s^{-1} and $\alpha_{2,r} = 51.02$ s^{-1}. Therefore, the complex angular frequencies $\alpha_2^* = \alpha_{2,r} + i2\pi f$ are known. The measured non-dimensional complex angular frequency Ω_2 one formulates as $\alpha_2^*/\alpha_{2,0}^*$, where $\alpha_{2,0}^*$ is the dimensional Rayleigh frequency. The results with the corresponding Ohnesorge numbers are shown by the black circles in figure 2. The error bars —barely visible in the diagrams—represent uncertainties in the frequency and decay rate of ±2 and ±11%, respectively, following the work by Brenn & Plohl [30]. The comparison yields good agreement between the measured and the calculated (solid black line) non-dimensional complex angular frequencies for the mode of initial deformation $m = 2$. For drops with different Oh_0, but the same $m = 2$, as well as De_1 and De_2, therefore, the values of Ω_2 can be taken from those lines. For the second-order approximation, the complex conjugate solution for $m = 4$ in figure 2 will be needed. This will be detailed later.

Since there exist pairs of solutions of the characteristic equation, the first-order solutions are rewritten, denoting the frequency with the positive imaginary part as $\alpha_m^{(p)}$, and the one with the negative imaginary part as $\alpha_m^{(n)}$. The resulting first-order deformation of the drop surface reads

$$\eta_1(\theta, t) = \left(\hat{\eta}_1^{(p)} e^{-\alpha_m^{(p)} t} + \hat{\eta}_1^{(n)} e^{-\alpha_m^{(n)} t}\right) P_m(\cos\theta)\,. \tag{3.13}$$

This formulation allows the two first-order initial conditions to be used for determining the amplitudes $\hat{\eta}_1^{(p)}$ and $\hat{\eta}_1^{(n)}$. We obtain

$$\hat{\eta}_1^{(p)} = -\frac{\alpha_m^{(n)}}{\alpha_m^{(p)} - \alpha_m^{(n)}} \quad \text{and} \quad \hat{\eta}_1^{(n)} = \frac{\alpha_m^{(p)}}{\alpha_m^{(p)} - \alpha_m^{(n)}}\,. \tag{3.14}$$

The first-order stream function, velocity components and pressure are given in the electronic supplementary materials. With these findings, the first-order solutions, which are formally identical to the Newtonian ones obtained by Zrnić *et al.* [36], are complete. One important difference is the modified Ohnesorge number entering the solutions, which depends on the frequency and on the two viscoelastic timescales.

(c) Second-order solutions

The second-order solutions are developed starting from pressure. We search for the general solution for the second-order pressure as a sum of two contributions

$$p_2(r, \theta, t) = p_{21}(r, \theta, t) + p_{22}(r, \theta, t), \tag{3.15}$$

where the subscript '21' denotes the solution of the inhomogeneous system of second-order equations, and the subscript '22' is the solution of the homogeneous system [36].

The solution with subscript '21' is first determined. We analyse the second-order equations of motion (2.23)–(2.25) and form the divergence of the vectorial momentum equation, which, with the use of the continuity equation (2.23), eliminates the second-order velocities from the momentum equation. The differential equation for the second-order pressure p_{21} is obtained as

$$\Delta p_{21} = -\nabla\cdot((\vec{u}_1\cdot\nabla)\vec{u}_1) + Oh_o \nabla\cdot[\nabla\cdot\boldsymbol{\tau}_{21}], \tag{3.16}$$

which we rewrite, using the Lamé identity, into the form

$$\Delta p_{21} = -\nabla \cdot \left[\nabla \vec{u}_1^{\,2}/2 - \vec{u}_1 \times \left(\nabla \times \vec{u}_1\right) - Oh_0 \nabla \cdot \boldsymbol{\tau}_{21}\right]. \tag{3.17}$$

The present flow field allows equation (3.17) to be rewritten as

$$\Delta\left[p_{21} + \vec{u}_1^{\,2}/2\right] = \nabla \cdot \left[\frac{1}{r^2 \sin^2\theta} \frac{1}{\beta_1 Oh_0} \frac{\partial \psi_2}{\partial t} \nabla \psi + Oh_0 \nabla \cdot \boldsymbol{\tau}_{21}\right]. \tag{3.18}$$

This is a Poisson equation for the modified pressure $\mathcal{P}_{21} = p_{21} + \vec{v}_1^{\,2}/2$ [36,46]. We determine the structure of the dependencies on the polar angular coordinate and time by inspecting the square of the first-order velocity vector, $\vec{v}_1^{\,2}$, which appears in the modified pressure. The inspected structure of the dependencies follows also from the products of first-order terms on the right-hand side of equation (3.18). Both groups of terms contain functions $\exp(-2\alpha_m t)$ for only positive (p) or negative (n) time dependency. The RCE (2.26) corresponding to the time dependencies according to $\exp(-2\alpha_m^{(p)} t)$ or $\exp(-2\alpha_m^{(n)} t)$, with the first-order extra stress tensor τ_1 known, reads

$$\boldsymbol{\tau}_{21} = 2\beta_2 \boldsymbol{D}_{21} + 2\gamma_1\left[\left(\vec{v}_1 \cdot \vec{\nabla}\right)\boldsymbol{D}_1 - \vec{\nabla}\vec{v}_1 \cdot \boldsymbol{D}_1 - \boldsymbol{D}_1 \cdot \vec{\nabla}\vec{v}_1^{\,T}\right]. \tag{3.19}$$

The coefficients in the equation are

$$\beta_2 = \frac{1 - 2\alpha_m De_2}{1 - 2\alpha_m De_1} \quad \text{and} \quad \gamma_1 = \frac{De_2 - \beta_1 De_1}{1 - 2\alpha_m De_1}. \tag{3.20}$$

For the positive (p) or negative (n) time dependency, we substitute the second-order extra stress tensor from equation (3.19) into the right-hand side of equation (3.18) and obtain

$$\Delta \mathcal{P}_{21} = \nabla \cdot \left[\frac{1}{r^2 \sin^2\theta} \frac{1}{\beta_1 Oh_0} \frac{\partial \psi_2}{\partial t} \nabla \psi + 2\gamma_1 Oh_0 \nabla \cdot \left[\left(\vec{v}_1 \cdot \vec{\nabla}\right)\boldsymbol{D}_1 - \vec{\nabla}\vec{v}_1 \cdot \boldsymbol{D}_1 - \boldsymbol{D}_1 \cdot \vec{\nabla}\vec{v}_1^{\,T}\right]\right], \tag{3.21}$$

since $\nabla \cdot (\nabla \cdot \boldsymbol{D}_{21}) = 0$. We substitute the stream function ψ and its constituent ψ_2 into the right-hand side of equation (3.21) and obtain

$$\begin{aligned}\Delta \mathcal{P}_{21} = &-q^2 C_{2m}{}^2 \Bigg\{\left[\frac{q^2}{(qr)^2}\left(\frac{d(qr\,j_m)}{dr}\right)^2 - \frac{q^4}{(qr)^2}(qr\,j_m)^2 + \right. \\ &\left. + \frac{C_{1m}}{C_{2m}}(m+1) r^{m-2} \frac{d(qr\,j_m)}{dr}\right](1-x^2)\left(\frac{dP_m(x)}{dx}\right)^2 + \\ &+ \left[\frac{q^4}{(qr)^4}(qr\,j_m)^2 + \frac{C_{1m}}{C_{2m}} r^{m-3}(qr\,j_m)\right](m(m+1)P_m(x))^2\Bigg\} e^{-2\alpha_m t} + \\ &+ \nabla \cdot \left[2\gamma_1 Oh_0 \nabla \cdot \left(\left(\vec{v}_1 \cdot \vec{\nabla}\right)\boldsymbol{D}_1 - \vec{\nabla}\vec{v}_1 \cdot \boldsymbol{D}_1 - \boldsymbol{D}_1 \cdot \vec{\nabla}\vec{v}_1^{\,T}\right)\right],\end{aligned} \tag{3.22}$$

where $x = \cos\theta$. In equation (3.22), the coefficients C_{1m} and C_{2m} and the quantity q represent either $\alpha_m^{(p)}$ or $\alpha_m^{(n)}$, which correspond to the time dependencies with $\exp(-2\alpha_m^{(p)} t)$ or $\exp(-2\alpha_m^{(n)} t)$, respectively. One further time dependency comes from products of terms with $\exp(-\alpha_m^{(p)} t)$ and $\exp(-\alpha_m^{(n)} t)$, which infer the time dependency with $\exp[-(\alpha_m^{(p)} + \alpha_m^{(n)})t]$. The RCE for this time dependency is obtained as

$$\begin{aligned}\boldsymbol{\tau}_{21}^{(pn)} = 2\beta_2^{(pn)} \boldsymbol{D}_{21}^{(pn)} &+ 2\gamma_1^{(pn)}\left[\left(\vec{v}_1^{\,(n)} \cdot \vec{\nabla}\right)\boldsymbol{D}_1^{(p)} - \vec{\nabla}\vec{v}_1^{\,(n)} \cdot \boldsymbol{D}_1^{(p)} - \boldsymbol{D}_1^{(p)} \cdot \vec{\nabla}\vec{v}_1^{\,(n)^T}\right] + \\ &+ 2\gamma_1^{(np)}\left[\left(\vec{v}_1^{\,(p)} \cdot \vec{\nabla}\right)\boldsymbol{D}_1^{(n)} - \vec{\nabla}\vec{v}_1^{\,(p)} \cdot \boldsymbol{D}_1^{(n)} - \boldsymbol{D}_1^{(n)} \cdot \vec{\nabla}\vec{v}_1^{\,(p)^T}\right],\end{aligned} \tag{3.23}$$

where the coefficient $\beta_2^{(pn)}$ is defined as

$$\beta_2^{(pn)} = \frac{1-\left(\alpha_m^{(p)}+\alpha_m^{(n)}\right)De_2}{1-\left(\alpha_m^{(p)}+\alpha_m^{(n)}\right)De_1} \tag{3.24}$$

and

$$\gamma_1^{(pn)} = \frac{De_2-\beta_1^{(p)}De_1}{1-\left(\alpha_m^{(p)}+\alpha_m^{(n)}\right)De_1}; \quad \gamma_1^{(np)} = \frac{De_2-\beta_1^{(n)}De_1}{1-\left(\alpha_m^{(p)}+\alpha_m^{(n)}\right)De_1}. \tag{3.25}$$

Substituting the second-order extra stress tensor from (3.23) into the right-hand side of equation (3.18), the equation for the time dependency with $\exp[-(\alpha_m^{(p)}+\alpha_m^{(n)})t]$ reads

$$\begin{aligned}\Delta\mathcal{P}_{21}^{(pn)} = \nabla\cdot\Bigg[&\frac{1}{r^2\sin^2\theta}\left(\frac{1}{\beta_1^{(p)}Oh_0}\frac{\partial\psi_2^{(p)}}{\partial t}\nabla\psi^{(n)}+\frac{1}{\beta_1^{(n)}Oh_0}\frac{\partial\psi_2^{(n)}}{\partial t}\nabla\psi^{(p)}\right)+\\ &+2\gamma_1^{(pn)}Oh_0\nabla\cdot\left[\left(\vec{v}_1^{(n)}\cdot\vec{\nabla}\right)\boldsymbol{D}_1^{(p)}-\vec{\nabla}\vec{v}_1^{(n)}\cdot\boldsymbol{D}_1^{(p)}-\boldsymbol{D}_1^{(p)}\cdot\vec{\nabla}\vec{v}_1^{(n)T}\right]+\\ &+2\gamma_1^{(np)}Oh_0\nabla\cdot\left[\left(\vec{v}_1^{(p)}\cdot\vec{\nabla}\right)\boldsymbol{D}_1^{(n)}-\vec{\nabla}\vec{v}_1^{(p)}\cdot\boldsymbol{D}_1^{(n)}-\boldsymbol{D}_1^{(n)}\cdot\vec{\nabla}\vec{v}_1^{(p)T}\right]\Bigg].\end{aligned} \tag{3.26}$$

We substitute the stream function ψ and its constituent ψ_2 into the right-hand side of equation (3.26) and obtain

$$\begin{aligned}\Delta\mathcal{P}_{21}^{(pn)} = -\Bigg\{\Bigg[&C_{2m}^{(p)}C_{2m}^{(n)}\frac{q^{(p)2}+q^{(n)2}}{r^2}\frac{d(q^{(p)}rj_m^{(p)})}{dr}\frac{d(q^{(n)}rj_m^{(n)})}{dr}\\ &-2C_{2m}^{(p)}C_{2m}^{(n)}\frac{q^{(p)2}q^{(n)2}}{r^2}(q^{(p)}rj_m^{(p)})(q^{(n)}rj_m^{(n)})+C_{1m}^{(n)}C_{2m}^{(p)}(m+1)q^{(p)2}r^{m-2}\frac{d(q^{(p)}rj_m^{(p)})}{dr}+\\ &+C_{1m}^{(p)}C_{2m}^{(n)}(m+1)q^{(n)2}r^{m-2}\frac{d(q^{(n)}rj_m^{(n)})}{dr}\Bigg]\left(\frac{dP_m(x)}{dx}\right)^2(1-x^2)+\\ &+\Bigg[C_{2m}^{(p)}C_{2m}^{(n)}\frac{q^{(p)2}+q^{(n)2}}{r^4}\left(q^{(p)}rj_m^{(p)}\right)\left(q^{(n)}rj_m^{(n)}\right)+C_{1m}^{(n)}C_{2m}^{(p)}q^{(p)2}r^{m-3}\left(q^{(p)}rj_m^{(p)}\right)+\\ &+C_{1m}^{(p)}C_{2m}^{(n)}q^{(n)2}r^{m-3}\left(q^{(n)}rj_m^{(n)}\right)\Bigg]\left(m(m+1)P_m(x)\right)^2\Bigg\}e^{-(\alpha_m^{(p)}+\alpha_m^{(n)})t}+\\ &+\nabla\cdot\Bigg\{\nabla\cdot\Bigg[2\gamma_1^{(pn)}Oh_0\left(\left(\vec{v}_1^{(n)}\cdot\vec{\nabla}\right)\boldsymbol{D}_1^{(p)}-\vec{\nabla}\vec{v}_1^{(n)}\cdot\boldsymbol{D}_1^{(p)}-\boldsymbol{D}_1^{(p)}\cdot\vec{\nabla}\vec{v}_1^{(n)T}\right)+\\ &+2\gamma_1^{(np)}Oh_0\left(\left(\vec{v}_1^{(p)}\cdot\vec{\nabla}\right)\boldsymbol{D}_1^{(n)}-\vec{\nabla}\vec{v}_1^{(p)}\cdot\boldsymbol{D}_1^{(n)}-\boldsymbol{D}_1^{(n)}\cdot\vec{\nabla}\vec{v}_1^{(p)T}\right)\Bigg]\Bigg\}.\end{aligned} \tag{3.27}$$

The further development of the right-hand sides of the Poisson equations for the modified pressure for all time dependencies is detailed in the electronic supplementary material owing to lengthy expressions.

The solution $\mathcal{P}_{21,H}$ of the Laplace equation for the modified pressure is first determined, which, for the time dependency according to $\exp(-2\alpha_m^{(p)}t)$, reads

$$\mathcal{P}_{21,H} = -\sum_{l=0}^{L} D_{21l}^{(p)} r^l P_l(\cos\theta) e^{-2\alpha_m^{(p)}t}. \tag{3.28}$$

In this equation, the $D_{21l}^{(p)}$ are expansion coefficients. The sums of these solutions, arising from the application of the Legendre polynomial orthogonality, are taken up to a maximum summation index (upper-case letter). Particular solutions $\mathcal{P}_{21,I}(r,\theta,t)$ for the right-hand sides in equations (3.22) and (3.27) in the forms of known functions seem to be out of reach. We therefore approximate the right-hand sides, replacing the spherical Bessel functions and products of Legendre functions by their series expansions for a given value of the mode m of initial drop deformation. Their forms are presented in the electronic supplementary material.

The particular solutions are searched in the forms of power series of the radial coordinate and of series of Legendre polynomials as functions of the polar angle. The approximation accuracy increases with the order of approximation by the series expansions.

The solution for the modified pressure is composed of the sum of the general solution of the Laplace equation and a particular solution of the Poisson equation as

$$\mathcal{P}_{21}(r,\theta,t) = \mathcal{P}_{21,H}(r,\theta,t) + \mathcal{P}_{21,I_1}(r,\theta,t) + \mathcal{P}_{21,I_2}(r,\theta,t), \tag{3.29}$$

where the particular solution $\mathcal{P}_{21,I_1}(r,\theta,t)$ satisfies the Poisson equation (3.22) for the modified pressure with only the right-hand side term in curled brackets, while the particular solution $\mathcal{P}_{21,I_2}(r,\theta,t)$ satisfies the Poisson equation with only the right-hand side term with the divergence. Analogous solutions are sought for the Poisson equation (3.27).

The second-order pressure contribution p_{21} accounts for all three different time dependencies and reads

$$p_{21}(r,\theta,t) = p_{21}^{(p)}(r,\theta,t) + p_{21}^{(n)}(r,\theta,t) + p_{21}^{(pn)}(r,\theta,t). \tag{3.30}$$

We determine the contribution to the pressure solution for the time dependency (p) with the definition of the modified pressure $\mathcal{P}_{21} = p_{21} + \vec{v}_1^{\,2}/2$ and obtain

$$\begin{aligned} p_{21}^{(p)}(r,\theta,t) = &-e^{-2\alpha_m^{(p)}t}\Bigg\{\sum_{l=0}^{L} D_{21l}^{(p)} r^l P_l(x) - \Bigg[\sum_{k=0}^{N}\sum_{l=0}^{2m} G_{2kl}^{(p)}\left(q^{(p)}r\right)^{2k+2m+2} P_l(x) + \\ &+ \sum_{k=0}^{N}\sum_{l=0}^{2m-1} \delta_{kl}^{(p)}\left(q^{(p)}r\right)^{2k+2m} P_l(x) + \sum_{k=1}^{N} \delta_{k(2m)}^{(p)}\left(q^{(p)}r\right)^{2k+2m} P_{(2m)}(x)\Bigg]\Bigg\} + \\ &+ \mathcal{P}_{21,I_2}^{(p)}(r,\theta,t) - \frac{1}{2}\left(\hat{\eta}_1^{(p)}\right)^2 \alpha_m^{(p)2}\left[f_{r1}^{(p)2}(r) P_m(\cos\theta)^2 + f_{\theta 1}^{(p)2}(r) P_m^1(\cos\theta)^2\right] e^{-2\alpha_m^{(p)}t}. \end{aligned} \tag{3.31}$$

The contributions $p_{21}^{(n)}$ and $p_{21}^{(pn)}$ are given in the electronic supplementary material. Owing to large expressions, the contribution $\mathcal{P}_{21,I_2}$ for all time dependencies is also detailed in the electronic supplementary material. The coefficients in these pressure solutions for each time dependency are determined as presented in the electronic supplementary material. The second-order solution p_{21} consists of two parts: the first two terms—the one in curled brackets and the term $\mathcal{P}_{21,I_2}^{(p)}(r,\theta,t)$—are obtained by solving the Poisson equation for the modified pressure. The second part represents the dynamic pressure caused by the first-order velocity field.

The corresponding second-order velocities are determined from the pressure field, the RCE equation (3.19) and the equations of motion (2.23)–(2.25). We obtain the radial momentum equation, with the modified pressure $\mathcal{P}_{21}$ introduced for the time dependency (p), as

$$\begin{aligned} \frac{\partial^2}{\partial r^2}\left(r^2 u_{r21}^{(p)}\right) + \frac{2\alpha_m^{(p)} r^2}{Oh_0 \beta_2^{(p)}} u_{r21}^{(p)} + \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial u_{r21}^{(p)}}{\partial\theta}\right) = \frac{r^2}{Oh_0\beta_2^{(p)}}\Bigg(&\frac{\partial\mathcal{P}_{21,H}^{(p)}}{\partial r} + \frac{\partial\mathcal{P}_{21,I_1}^{(p)}}{\partial r} \\ &- \frac{1}{r^2\sin^2\theta}\frac{1}{\beta_1^{(p)} Oh_0}\frac{\partial\psi_2^{(p)}}{\partial t}\frac{\partial\psi^{(p)}}{\partial r} + \frac{\partial\mathcal{P}_{21,I_2}^{(p)}}{\partial r} - \frac{1}{r^2}\frac{\partial^2}{\partial r^2}\left(r^2 F_{rr21}^{(p)}\right) \\ &- \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}\left(F_{r\theta 21}^{(p)}\sin\theta\right) + \frac{F_{\theta\theta 21}^{(p)} + F_{\phi\phi 21}^{(p)}}{r}\Bigg), \end{aligned} \tag{3.32}$$

where the tensor

$$\boldsymbol{F}_{21}^{(p)} = 2\gamma_1^{(p)} Oh_0\left[\left(\vec{v}_1^{(p)}\cdot\vec{\nabla}\right)\boldsymbol{D}_1^{(p)} - \vec{\nabla}\vec{v}_1^{(p)}\cdot\boldsymbol{D}_1^{(p)} - \boldsymbol{D}_1^{(p)}\cdot\left(\vec{\nabla}\vec{v}_1^{(p)}\right)^T\right]. \tag{3.33}$$

The equation for the time dependency (n) is analogous. For the time dependency according to $\exp[-(\alpha_m^{(p)}+\alpha_m^{(n)})t]$, the radial momentum equation reads

$$\frac{\partial^2}{\partial r^2}(r^2u_{r21}^{(pn)})+\frac{\left(\alpha_m^{(p)}+\alpha_m^{(n)}\right)r^2}{Oh_0\beta_2^{(pn)}}u_{r21}^{(pn)}+\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial u_{r21}^{(pn)}}{\partial\theta}\right)=\frac{r^2}{Oh_0\beta_2^{(pn)}}\left(\frac{\partial\mathcal{P}_{21,H}^{(pn)}}{\partial r}+\right.$$
$$+\frac{\partial\mathcal{P}_{21,I_1}^{(pn)}}{\partial r}-\frac{1}{r^2\sin^2\theta}\left(\frac{1}{Oh_0\beta_1^{(p)}}\frac{\partial\psi_2^{(p)}}{\partial t}\frac{\partial\psi^{(n)}}{\partial r}+\frac{1}{Oh_0\beta_1^{(n)}}\frac{\partial\psi_2^{(n)}}{\partial t}\frac{\partial\psi^{(p)}}{\partial r}\right)+\frac{\partial\mathcal{P}_{21,I_2}^{(pn)}}{\partial r}$$
$$\left.-\frac{1}{r^2}\frac{\partial^2}{\partial r^2}\left(r^2F_{rr21}^{(pn)}\right)-\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}\left(F_{r\theta21}^{(pn)}\sin\theta\right)+\frac{F_{\theta\theta21}^{(pn)}+F_{\phi\phi21}^{(pn)}}{r}\right), \tag{3.34}$$

where

$$\boldsymbol{F}_{21}^{(pn)}=2\gamma_1^{(pn)}Oh_0\left[\left(\vec{v}_1^{(n)}\cdot\vec{\nabla}\right)\boldsymbol{D}_1^{(p)}-\vec{\nabla}\vec{v}_1^{(n)}\cdot\boldsymbol{D}_1^{(p)}-\boldsymbol{D}_1^{(p)}\cdot\vec{\nabla}\vec{v}_1^{(n)^T}\right]+$$
$$+2\gamma_1^{(np)}Oh_0\left[\left(\vec{v}_1^{(p)}\cdot\vec{\nabla}\right)\boldsymbol{D}_1^{(n)}-\vec{\nabla}\vec{v}_1^{(p)}\cdot\boldsymbol{D}_1^{(n)}-\boldsymbol{D}_1^{(n)}\cdot\vec{\nabla}\vec{v}_1^{(p)^T}\right]. \tag{3.35}$$

The further development of the second-order radial momentum equation for all the time dependencies is detailed in the electronic supplementary material. The equation is solved for all the time dependencies of the second-order radial velocity u_{r21} to obtain

$$u_{r21}(r,\theta,t)=u_{r21}^{(p)}(r,\theta,t)+u_{r21}^{(n)}(r,\theta,t)+u_{r21}^{(pn)}(r,\theta,t). \tag{3.36}$$

The solutions u_{r21} for all the time dependencies are given as the sum of the general solutions of the homogeneous differential equations and one particular solution of the inhomogeneous equations. As an example, we present the solution for the positive time dependency (p), which reads

$$u_{r21}^{(p)}(r,\theta,t)=-\sum_{l=0}^{L}\left[A_{21l}^{(p)}r^{l-1}+B_{21l}^{(p)}q_2^{(p)2}\frac{j_l(\sqrt{2}q_2^{(p)}r)}{q_2^{(p)}r}\right]l(l+1)P_l(x)e^{-2\alpha_m^{(p)}t}+$$
$$+\left\{\sum_{k=0}^{N}\sum_{l=0}^{2m}\left[K_{1kl}^{(p)}\left(q^{(p)}r\right)^{2k+2m+1}+K_{2kl}^{(p)}\left(q^{(p)}r\right)^{2k+2m+3}\right]P_l(x)+\right.$$
$$+\sum_{k=0}^{N}\sum_{l=0}^{2m-1}K_{3kl}^{(p)}\left(q^{(p)}r\right)^{2k+2m+1}P_l(x)+$$
$$\left.+\sum_{k=1}^{N}K_{3k(2m)}^{(p)}\left(q^{(p)}r\right)^{2k+2m+1}P_{(2m)}(x)\right\}e^{-2\alpha_m^{(p)}t}+Bp_{r21}^{(p)}(r,\theta,t)$$
$$=:-\sum_{l=0}^{L}\left(A_{21l}^{(p)}f_{ar}^{(p)}+B_{21l}^{(p)}f_{br}^{(p)}\right)P_l(\cos\theta)e^{-2\alpha_m^{(p)}t}+Ap_{r21}^{(p)}(r,\theta,t)+Bp_{r21}^{(p)}(r,\theta,t), \tag{3.37}$$

where we have introduced the new argument $q_2^{(p)}=\sqrt{\alpha_m^{(p)}/\beta_2^{(p)}Oh_0}$ of the spherical Bessel functions. The radial velocity components, together with the solutions $Ap_{r21}(r,\theta,t)$ and $Bp_{r21}(r,\theta,t)$ for every time dependency, are given and explained in the electronic supplementary material.

The known radial velocity component u_{r21} allows the second-order velocity component in the direction of the polar angle to be determined, using the continuity equation (2.23). The solution is represented as

$$u_{\theta21}(r,\theta,t)=u_{\theta21}^{(p)}(r,\theta,t)+u_{\theta21}^{(n)}(r,\theta,t)+u_{\theta21}^{(pn)}(r,\theta,t). \tag{3.38}$$

The solution for the positive time dependency (p) reads

$$
\begin{aligned}
u_{\theta 21}^{(p)}(r,\theta,t) = & -\Bigg\{\sum_{l=0}^{L}\Bigg[A_{21l}^{(p)}(l+1)r^{l-1} + \\
& + B_{21l}^{(p)}q_2^{(p)2}\left((l+1)\frac{j_l(\sqrt{2}q_2^{(p)}r)}{q_2^{(p)}r} - \sqrt{2}j_{l+1}\left(\sqrt{2}q_2^{(p)}r\right)\right)\Bigg]P_l^1(x) + \\
& + \sum_{k=0}^{N}\sum_{l=0}^{2m}\left[L_{1kl}^{(p)}\left(q^{(p)}r\right)^{2k+2m+1} + L_{2kl}^{(p)}\left(q^{(p)}r\right)^{2k+2m+3}\right]\frac{P_{l+1}(x)-P_{l-1}(x)}{\sqrt{1-x^2}} + \\
& + \sum_{k=0}^{N}\sum_{l=0}^{2m-1}L_{3kl}^{(p)}\left(q^{(p)}r\right)^{2k+2m+1}\frac{P_{l+1}(x)-P_{l-1}(x)}{\sqrt{1-x^2}} + \\
& + \sum_{k=1}^{N}L_{3k(2m)}^{(p)}\left(q^{(p)}r\right)^{2k+2m+1}\frac{P_{(2m+1)}(x)-P_{(2m-1)}(x)}{\sqrt{1-x^2}}\Bigg\}e^{-2\alpha_m^{(p)}t} + \\
& + Bp_{\theta 21}^{(p)}(r,\theta,t) =: -\sum_{l=0}^{L}\left(A_{21l}^{(p)}f_{a\theta}^{(p)} + B_{21l}^{(p)}f_{b\theta}^{(p)}\right)P_l^1(\cos\theta)e^{-2\alpha_m^{(p)}t} + \\
& + Ap_{\theta 21}^{(p)}(r,\theta,t) + Bp_{\theta 21}^{(p)}(r,\theta,t)\,.
\end{aligned}
\tag{3.39}
$$

The functions f_{ar}, $f_{a\theta}$, f_{br} and $f_{b\theta}$, defined for each oscillation frequency, are introduced to facilitate the expressions of the flow variables. The solutions for the polar angular velocity component for the two other time dependencies are given in the electronic supplementary material. The solutions $Ap_{\theta 21}^{(p)}(r,\theta,t)$ and $Bp_{\theta 21}^{(p)}(r,\theta,t)$, together with $Ap_{r21}^{(p)}(r,\theta,t)$ and $Bp_{r21}^{(p)}(r,\theta,t)$, respectively, satisfy the second-order continuity equation.

The structure of the above-given solutions determines the second-order deformation of the drop surface η_{21} as

$$
\eta_{21}(\theta,t) = \sum_{l=0}^{L}\left(H_{21l}^{(p)}e^{-2\alpha_m^{(p)}t} + H_{21l}^{(n)}e^{-2\alpha_m^{(n)}t} + H_{21l}^{(pn)}e^{-\left(\alpha_m^{(p)}+\alpha_m^{(n)}\right)t}\right)P_l(\cos\theta). \tag{3.40}
$$

The deformed shape is governed by all the Legendre polynomials in the sum. The reason is that, in the zero normal-stress boundary condition (2.29), pressure and surface deformation are coupled. The set of unknown coefficients A_{21l}, B_{21l} and H_{21l} are determined for each time dependency and summation index l from the second-order boundary conditions (2.27)–(2.29). The electronic supplementary material details the calculation of these coefficients.

The second contributions to the second-order solutions, with subscript '22', are determined from the homogeneous forms of equations (2.23), (2.24) and (2.25). This set of equations has the same structure as at first order, where the complex angular frequency α_m is replaced by α_{2k} (with the number '2' in the subscript indicating the second-order solution, and k a deformation mode number), and the deformation amplitude of the drop surface $\hat{\eta}_1$ by $\hat{\eta}_{22k}$. Both replaced variables are under a sum with the summation index k, implying that all possible deformation modes k affect the results.

For the exponential dependency on time via $\exp(-\alpha_{2k}t)$, positive (p) or negative (n), we find the second-order solutions of the extra-stress tensor, with subscript '22'

$$
\boldsymbol{\tau}_{22} = 2\sum_{k=0}^{K}\frac{1-De_2\alpha_{2k}}{1-De_1\alpha_{2k}}\boldsymbol{D}_{22k} =: 2\sum_{k=0}^{K}\beta_{2k}\boldsymbol{D}_{22k}\;. \tag{3.41}
$$

The second-order pressure, velocity and drop surface deformation '22' are linear combinations of eigensolutions of the differential equations with the summation index k. They exhibit forms close to the solutions of the first order. As an example, the velocity component u_{r22} in the radial direction reads

$$
\begin{aligned}
u_{r22}(r,\theta,t) = &- \sum_{k=0}^{K} \hat{\eta}_{22k}^{(p)} \alpha_{2k}^{(p)} e^{-\alpha_{2k}^{(p)} t} \left[r^{k-1} + \frac{2(k^2-1)}{2q_{2k}^{(p)} Q_k^{(p)} - q_{2k}^{(p)2}} \left(r^{k-1} - \frac{j_k(q_{2k}^{(p)} r)}{j_k(q_{2k}^{(p)})} \frac{1}{r} \right) \right] \times \\
&\times P_k(\cos\theta) - \sum_{k=0}^{K} \hat{\eta}_{22k}^{(n)} \alpha_{2k}^{(n)} e^{-\alpha_{2k}^{(n)} t} \left[r^{k-1} + \frac{2(k^2-1)}{2q_{2k}^{(n)} Q_k^{(n)} - q_{2k}^{(n)2}} \left(r^{k-1} - \frac{j_k(q_{2k}^{(n)} r)}{j_k(q_{2k}^{(n)})} \frac{1}{r} \right) \right] P_k(\cos\theta) = \\
&=: - \sum_{k=0}^{K} \left[\hat{\eta}_{22k}^{(p)} \alpha_{2k}^{(p)} e^{-\alpha_{2k}^{(p)} t} f_{r2}^{(p)}(r) + \hat{\eta}_{22k}^{(n)} \alpha_{2k}^{(n)} e^{-\alpha_{2k}^{(n)} t} f_{r2}^{(n)}(r) \right] P_k(\cos\theta),
\end{aligned} \tag{3.42}
$$

with the argument $q_{2k} = \sqrt{\alpha_{2k}/\beta_{2k} Oh_0}$ and the definition of the function $Q_k = j_{m+1}(q_{2k})/j_m(q_{2k})$. The function f_{r2} for each oscillation frequency is introduced in the interest of compact formulations. The solutions $u_{\theta 22}$ and p_{22} are given in the electronic supplementary material. The drop surface deformation '22' reads

$$
\eta_{22}(\theta,t) = \sum_{k=0}^{K} \left(\hat{\eta}_{22k}^{(p)} e^{-\alpha_{2k}^{(p)} t} + \hat{\eta}_{22k}^{(n)} e^{-\alpha_{2k}^{(n)} t} \right) P_k(\cos\,\theta). \tag{3.43}
$$

The characteristic equation obtained from the boundary condition (2.29) determines the complex conjugate angular frequencies α_{2k}. This equation is formally identical to the first-order equation (3.11), but it is solved by replacing the mode m of initial drop deformation by the summation index k, as required by the solution. The amplitudes $\hat{\eta}_{22k}^{(p)}$ and $\hat{\eta}_{22k}^{(n)}$ are calculated using the second-order initial conditions (2.30) [46]. The sum of the two contributions to the second-order drop deformation must satisfy the second-order initial conditions for every summation index k. These contributions therefore depend on each other. Applying the initial conditions, one obtains a set of two equations with the two unknown coefficients $\hat{\eta}_{22k}^{(p)}$ and $\hat{\eta}_{22k}^{(n)}$. The method applied for determining these coefficients is the same as on page 10 of our paper Zrnić *et al*. [36]. We therefore do not repeat its description here.

This analysis shows that the cases of a volumetric drop pulsation ($k=0$) and a translation motion of the drop ($k=1$) drop out from the solutions '22'. This is plausible, since the case $k=0$ is impossible for incompressible liquid, and the case $k=1$ would be possible only owing to a resultant force acting on the drop, which does not exist here.

The two amplitudes $\hat{\eta}_{22k}^{(p)}$ and $\hat{\eta}_{22k}^{(n)}$ for values $k \geq 2$ are determined using the second-order initial condition (2.30). The coefficient H_{21k} is known from the boundary conditions, and the equation for the amplitude $\hat{\eta}_{22k}^{(p)}$ is

$$
\hat{\eta}_{22k}^{(p)} = \frac{1}{\alpha_{2k}^{(p)} - \alpha_{2k}^{(n)}} \left[\alpha_{2k}^{(n)} \left(H_{21k}^{(p)} + H_{21k}^{(n)} + H_{21k}^{(pn)} \right) - \left(2\alpha_m^{(p)} H_{21k}^{(p)} + 2\alpha_m^{(n)} H_{21k}^{(n)} + (\alpha_m^{(p)} + \alpha_m^{(n)}) H_{21k}^{(pn)} \right) \right]. \tag{3.44}
$$

The equation for the amplitude $\hat{\eta}_{22k}^{(n)}$ is analogous. Having found the complete second-order solutions, in the following section we will present and validate various results for viscoelastic liquid drop-shape oscillations.

4. Results and discussion

In this section, we evaluate the solutions derived above. First, we verify the conservation of the drop volume for the two drops of table 1, varying the deformation parameter η_0. Then we present traces of the drop aspect ratio and deformed drop shapes for different viscoelastic liquid material properties and compare the results on the nonlinear behaviour with data for a corresponding viscous case with the mode of initial deformation $m=2$. We then show two well-known nonlinear effects in drop-shape oscillations, which are the excess time in the drop prolate form and the frequency change with varying initial deformation. For selected traces of the drop north pole motion, we analyse the quasi-periodicity of motion and the Fourier power spectra. We conclude by a study of the onset of aperiodic behaviour.

(a) Volume conservation

First, we verify the conservation of the drop volume, because the weakly nonlinear theory represents all the field variables, together with the drop surface shape, as truncated power series of a small deformation parameter η_0. Consequently, the non-dimensional drop volume may deviate from its exact value of $V_s = 4\pi/3$. The deviation is determined analytically as a function of time and reads

$$\frac{V(t) - V_s}{V_s} = \frac{1}{2}\int_{-1}^{1}\left[r_s^3(\theta, t) - 1\right]d\cos\theta =: R(\eta_0, t), \tag{4.1}$$

where the non-dimensional drop volume $V(t)$ is determined as

$$V(t) = \frac{2\pi}{3}\int_{-1}^{1} r_s^3(\theta, t)d\cos\theta = \frac{2\pi}{3}\int_{-1}^{1}\left[1 + \eta_1(\theta, t)\eta_0 + \eta_2(\theta, t)\eta_0^2\right]^3 d\cos\theta. \tag{4.2}$$

The values of R are all positive, oscillate in time and increase with η_0. Figure 4 presents the results at the instant of time $t = t_{max}$ of largest volume deviation for the mode of initial deformation $m = 2$ for the two viscoelastic drops in table 1. In both cases, for deformation parameters $\eta_0 \leq 0.35$, the volume deviates by less than 2% from the exact value. This is considered acceptable for the present analysis.

(b) Deformed drop shapes

Figure 5 presents the drop aspect ratio $r_s(0, t)/r_s(\pi/2, t)$ as a function of time for the mode of initial deformation $m = 2$ for three different drops: the two viscoelastic liquid drops from table 1, with $Oh_0 = 0.16$ and $Oh_0 = 1.238$, and a viscous drop with $Oh = 0.048$. The value of the viscous drop Ohnesorge number is selected such that the linear value of its decay rate is equal to that value for the viscoelastic drop with $Oh_0 = 1.238$. For this comparison, all the results presented account for the solutions up to the second order of approximation. For the viscoelastic drop with $Oh_0 = 1.238$, the Ohnesorge number is well beyond the critical value of 0.7665 of the Newtonian case. In that case, the result of the characteristic equation suggests that such drops have an aperiodic response to the initial deformation. This is not the case, however, for the far higher Ohnesorge number $Oh_0 = 1.238$ in the viscoelastic case, which figure 5 confirms, showing shape oscillations of the drop. The driver of this viscoelastic drop oscillation is the elasticity, in contrast to the viscous case, where it is surface tension. Another interesting comparison below the critical value is between the viscous drop with $Oh = 0.048$ and the viscoelastic drop with $Oh_0 = 1.238$. In these two cases, the linear values of the decay rate are the same, but initially figure 5 shows significantly different peaks of the drop aspect ratio between the two cases. The difference can be explained by the involved elasticity, which plays a significant role in large deformations, raises the maximum amplitudes and changes the resultant decay rate and frequency of the oscillation in comparison with the viscous case. The difference in the minimum aspect ratio in the first oblate state between these two drops is around 12%. For the two compared drops, the minimum and maximum peaks of the aspect ratio are more pronounced for the viscoelastic drop than for the viscous drop throughout the oscillations. This behaviour is explained by the elasticity as an extra driving force. As a further consequence, these peaks always appear earlier in time than for the viscous drop, i.e. the oscillation frequency of the viscoelastic drop is higher.

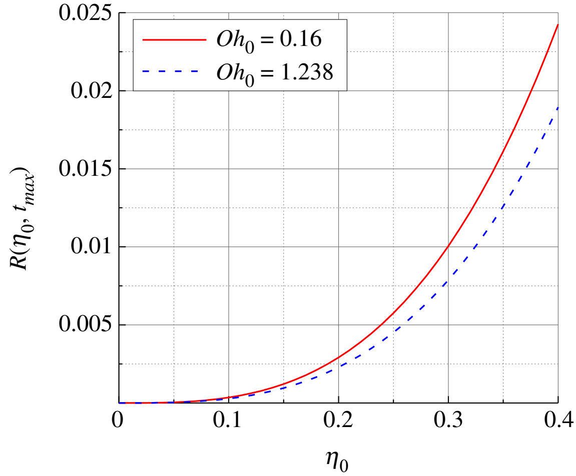

Figure 4. Maximum relative volume deviation from the exact value for the viscoelastic liquid drops from table 1 with $Oh_0 = 0.16$ and $Oh_0 = 1.238$ at $m = 2$ as a function of the deformation parameter η_0.

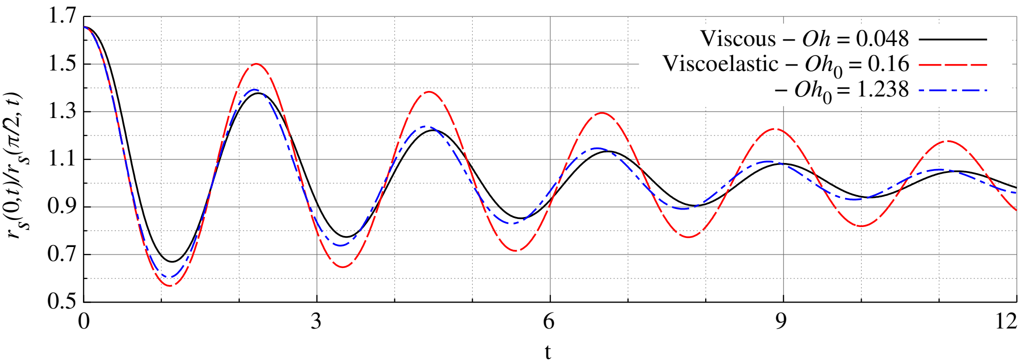

Figure 5. The drop aspect ratio over time for three liquid drops with $m = 2$, $\eta_0 = 0.35$, from the theory up to the second order.

The present results from the analysis allow the deformed drop shapes up to second order to be plotted, representing nonlinear effects. Figure 6*a,b* show the meridional sections of deformed drops for the mode of initial drop deformation $m = 2$, i.e. for prolate-to-oblate oscillations, with the relatively large initial aspect ratio of the drop of $L/W = 1.65$ of the three liquid drops in figure 5. The degrees of Legendre polynomials contributing to the solutions of the second order are 0, 2 and 4. The drop shapes are represented for the first oscillation period at the states of maximum deformation, omitting the initial state. The time instants of the maximum states shown in figure 6*a,b* are related to the trace of the north pole data in figure 5 for $Oh_0 = 1.238$. The corresponding case assumes the maximum deformed states at the earliest times in comparison with the drops with $Oh = 0.048$ and $Oh_0 = 0.16$. Owing to the dampening, the first oblate and second prolate shapes are, of course, less deformed than the initial one. With ongoing time, as oscillation amplitudes decrease, the shapes converge to the linear ones.

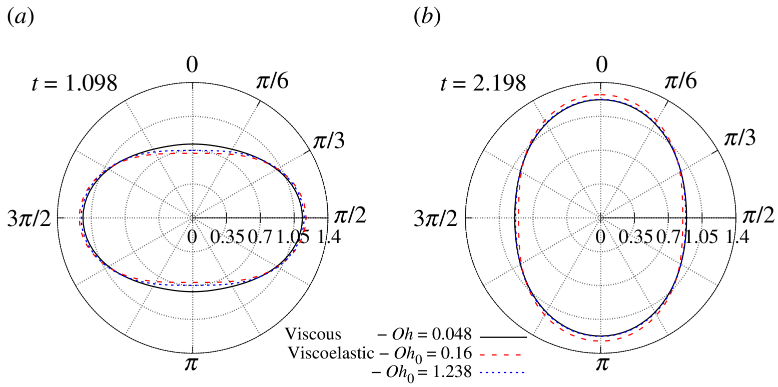

Figure 6. The (*a*) first oblate and (*b*) second prolate shapes for the initial deformation mode $m = 2$ with $\eta_0 = 0.35$ of three liquid drops. The shapes are axisymmetric around the vertical axis ($0 - \pi$). The time instants in (*a*) and (*b*) are chosen for the first minimum and second maximum drop aspect ratio, seen in figure 5, for the viscoelastic drop with $Oh_0 = 1.238$. The results include solutions up to the second order.

(c) Timescales of the drop-shape oscillations

Two well-known nonlinear effects in drop-shape oscillations with $m = 2$ are the asymmetry of the times the drop spends in the oblate and prolate deformed states and a frequency change with varying drop deformation. One expects the former effect to be less pronounced in the present viscoelastic than in the inviscid case, but similar to the viscous case. The second-order solutions can represent these phenomena. In figure 7*a*, the second-order solutions for $m = 2$ show the time the drop spends in the elongated form, determined from the first oscillation, except for the inviscid analysis that is based on the solutions up to third order and data are based on the first 10 oscillations [35]. The shaded area presents the s.d. from the mean value denoted with the solid black line. The relatively high Ohnesorge number of 0.1 yields only a few oscillations, whereas the second and following periods exhibit very small amplitudes and thus do not show nonlinearities. The relatively small Ohnesorge number of 0.048 shows more pronounced excess time and in comparison with other viscous and viscoelastic liquid drop cases presents a case with the strongest nonlinear effects. In the comparison of the two viscoelastic cases with $Oh_0 = 0.16$ and $Oh_0 = 1.238$, we see a difference, but not as significant as in the comparison between the two viscous cases, although the Ohnesorge number difference is smaller. Also, both viscoelastic cases yield significantly smaller excess times for the present range of initial drop aspect ratios. Comparing the viscoelastic drop with $Oh_0 = 1.238$ and the viscous drop with $Oh = 0.048$, which have the same linear decay rate, the latter drop has more than twice as large excess time than the former for the whole range of initial aspect ratios studied. This means that the elasticity reduces the excess time in favour of an increase in the maximum amplitudes of the drop north pole motion.

Figure 7*b* shows the frequency change of the drop north pole motion for the initial deformation mode $m = 2$ as a function of the initial drop aspect ratio. The results are presented in the same way as in figure 7*a*. The frequency change is normalized by the imaginary part of the first-order oscillation frequency $\alpha_{2,i}$ for the viscous and viscoelastic cases. In the inviscid case also shown, the frequency change is normalized by the Rayleigh frequency. The results by Zrnić & Brenn [35] for the inviscid case are shown to quantify the difference between the inviscid on the one hand and the viscous and viscoelastic drops on the other. We do not see a significant change in frequency for the second-order solutions, which suggests that an analysis up to the third order of approximation could add to these results.

The next subject of our study is the oscillatory motion of the drop north pole for the mode of initial deformation $m = 2$. Figure 8*a* depicts the drop north pole position for the deformation

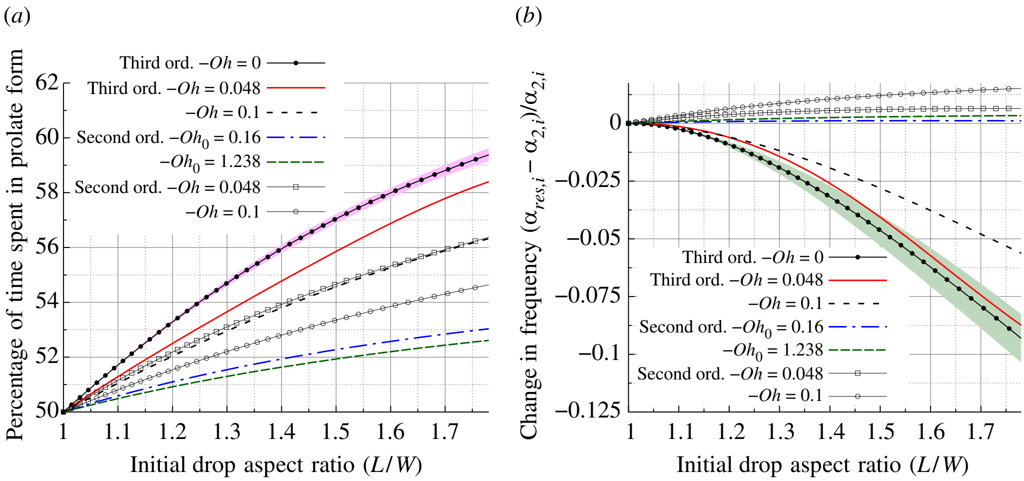

Figure 7. (*a*) Percentage of the oscillation period spent in prolate form and (*b*) change in frequency as functions of the initial drop aspect ratio L/W for $m = 2$. The maximum deformation parameter $\eta_0 = 0.4$ corresponds to the maximum value on the *x*-axis $L/W = 1.78$.

parameter $\eta_0 = 0.35$ as a function of time for the same three liquid drops: the two viscoelastic liquid drops from table 1, and the viscous drop with $Oh = 0.048$. The first four periods are denoted by $t_{p,1}$–$t_{p,4}$ in figure 8*a*. These oscillation periods are evaluated for the whole of the two oscillations in figure 8*a*. The viscoelastic results from this analysis show that, with ongoing time, the period length changes, leading to a slight decrease in the frequency in time. Accounting for the third-order solution in the viscous case leads to an increase in the frequency in time [36]. These behaviours indicate a time dependency of the frequency for nonlinear drop oscillations, as stated by Prosperetti [8] for the Newtonian case. The data in figure 8*b* show a non-monotonic convergence of the oscillation periods towards the linear values. For the Ohnesorge numbers $Oh = 0.16$ and $Oh = 1.238$, the linear theory predicts the non-dimensional oscillation periods of 2.222 and 2.202, respectively. These values are depicted by the red dashed and blue dotted horizontal lines in the figure. At the start of the motion, for $Oh_0 = 0.16$ and $Oh_0 = 1.238$ with the larger deformation parameter $\eta_0 = 0.35$, the period is shorter than the linear value by −0.59 and −0.68%, respectively. Another interesting finding is that the beat between different modes of oscillation may make the period length suddenly increase, even during cycles with small deformations. The quasi-periodicity of the drop-shape oscillations is caused by a beat between various oscillation modes excited in the course of the motion [35]. This beat phenomenon, however, is dampened as the viscosity increases, as shown in figure 8*b*, where the data for the higher Ohnesorge number fluctuate much less than those for the lower one.

Next, we determine the damping rate α_r from the results for the oscillations depicted in figure 8*a*. The damping rate is given by pairs of data points representing states of maximum positive or negative deformation against the spherical shape. These points define an oscillation period. Examples of such oscillations are marked by the starting and ending times of the periods $t_{p,1}$ and $t_{p,3}$ in figure 8*a*. For every such pair of drop north pole positions at an earlier time t_1 and a later time t_2, the decay rate is deduced as $\alpha_r = \ln[\eta(0, t_2)/\eta(0, t_1)]/(t_1 - t_2)$. The decay rate is plotted in figure 8*c* against the starting time of an oscillation for the three different liquid drops. The viscoelastic drop with $Oh_0 = 0.16$ converges to the value of 0.109, which is the real part of the complex angular frequency α_2 predicted by the linear theory, as depicted by the red dashed horizontal line in figure 8*c*. The damping rate for the viscous and viscoelastic drops with $Oh = 0.048$ and $Oh_0 = 1.238$, respectively, converges in time to the blue dotted horizontal line at the value of 0.21, which is the real part of the complex angular frequency α_2

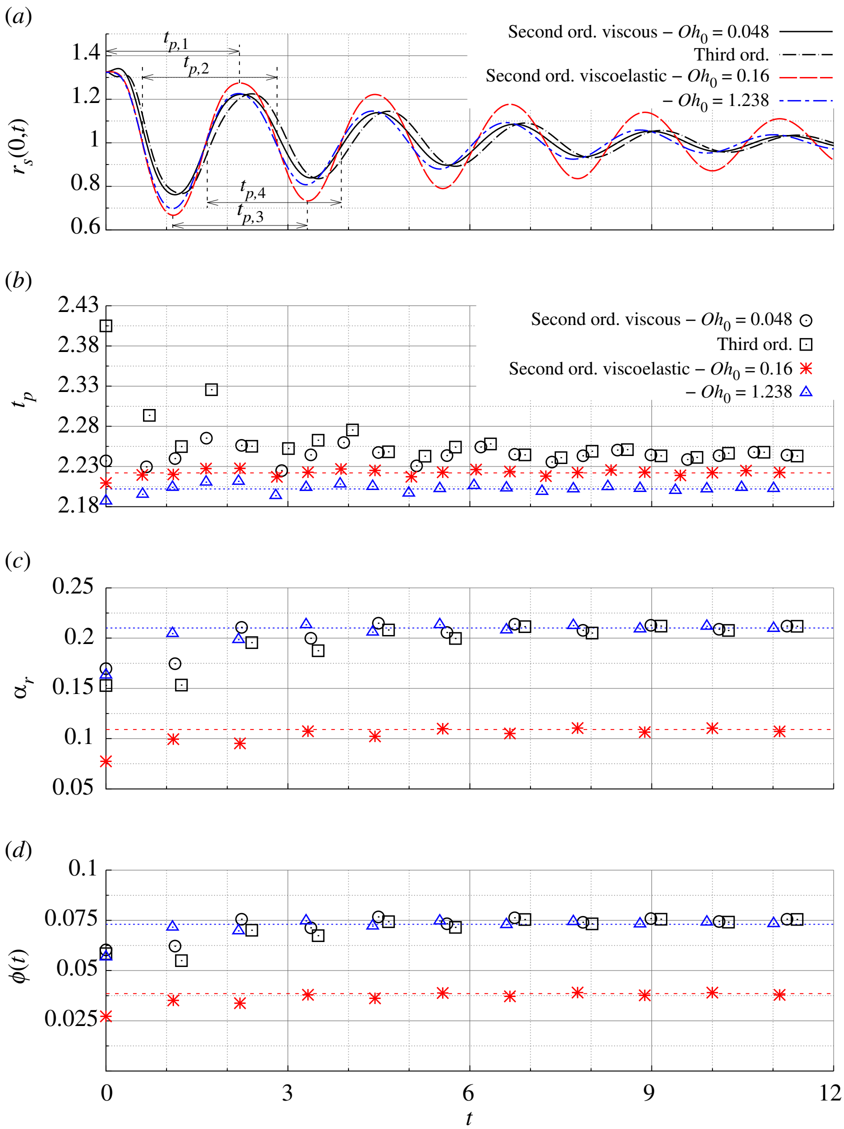

Figure 8. (*a*) Position of the drop north pole for three different liquid drops with the deformation parameter $\eta_0 = 0.35$. Double arrows mark periods t_p starting from different time instants. (*b*) Periods t_p measured from different starting times during the oscillation are shown in (*a*). (*c*) Damping rate is determined for different pairs of displacement maxima or minima. (*d*) Loss angle $\phi(t)$. Mode of initial deformation $m = 2$.

for $Oh_0 = 1.238$ predicted by the linear theory. Comparing the phenomenon between the two viscoelastic liquid drops, we conclude that, in nonlinear drop oscillations, viscosity affects the damping rate differently. The larger viscosity does not necessarily dampen the system more rapidly, so that, at $\eta_0 = 0.35$, the drop with $Oh_0 = 1.238$ approaches the linear value similarly fast as the drop with $Oh_0 = 0.16$. For the former and latter viscoelastic liquid drops, the damping rate from the first period $t_{p,1}$ deviates around −22.4 and −29.3%, respectively, from the linear value. The viscoelastic liquid drop with $Oh_0 = 1.238$ is dampened faster than the viscous drop with the same decay rate at first order. The values of the decay rate extracted from the first oscillation period $t_{p,1}$, using up to the second-order solution, are similar, assuming 0.163 and

0.17 for a viscoelastic drop with $Oh_0 = 1.238$ and the viscous drop with $Oh_0 = 0.048$, respectively. In all cases, the return of a nonlinear damped to a linear motion with ongoing time is seen, since the deformations decrease in time. The trend of the decay rate to increase in time agrees with the effect found for Newtonian drop-shape oscillations by Zrnić *et al.* [36], which is seen in the Newtonian data in figure 8*c* again. With ongoing time, the data converge to the linear respective value for the given drop liquid, where the values for $Oh = 0.048$ and for $Oh_0 = 1.238$ are the same, since the Newtonian drop with $Oh = 0.048$ was so designed. It is interesting to see that the decay rates of the viscoelastic drop oscillations converge faster to the linear values than the Newtonian. The effect of elasticity is the explanation for this behaviour.

The viscous influence on the viscoelastic drop-shape oscillations is reflected by the loss angle $\phi(t) = \arctan(\alpha_r/\alpha_i)$ of the motion also, as depicted in figure 8*d*. For the viscoelastic drops, the angle is higher for the higher Ohnesorge number Oh_0, and, analogously to the decay rate, it converges faster to the linear values for the viscoelastic than for the Newtonian drops. The linear values are 0.0385 and 0.073 for $Oh_0 = 0.16$ and 1.238, respectively. The loss angles in the first oscillation period $t_{p,1}$ for the former and the latter Oh_0 deviate from the respective linear values by 29.3 and 21.9%, respectively.

(d) Power spectra

For the drop-shape oscillations studied, we are interested in the oscillation frequencies contributing to the motions. For quantifying this, we compute the Fourier power spectra of the frequency involved in the data traces in time. The finite-time frequency spectrum of a function of time $g(t)$ is given in an analytical form as

$$\hat{g}(\alpha) = \int_{t_1}^{t_2} g(t) e^{-i\alpha t} dt. \tag{4.3}$$

The integral is defined for the time interval between t_1 and t_2. The Fourier transform becomes more important for finding the frequencies, the more the quasi-periodicity influences the time series in motions with large deformations. The quasi-periodicity depends on the liquid drop material properties also. The frequencies are then not easily found from a 'manual' inspection of the time series. Figure 9 shows the Fourier power spectra of the frequency for the motions of the drop north pole in figure 8*a*. The data present power densities for the three different liquid drops with the deformation parameter $\eta_0 = 0.35$. The spectra exhibit zero values at zero frequency because the underlying data represent the deviations of the north pole position from its time average. The spectra are shown as obtained from an analysis of the first oscillation period only and from analysing the period of time $0 \le t \le 12$ in order to see the evolution of the frequency spectra with ongoing time during the oscillations.

For the viscous and viscoelastic oscillating drops with $Oh = 0.048$ and $Oh_0 = 1.238$ studied, the method yields similar power spectra of the oscillation frequency, as represented in the diagrams by the solid black and dotted blue lines, respectively. The spectra of the first oscillations exhibit peaks at frequencies deviating from the solutions of the characteristic equation of the drop, i.e. from the linear oscillation frequency. The complex angular frequencies, which represent the solutions of the characteristic equation, are $0.109 \pm 2.827i$ and $0.21 \pm 2.853i$ for the viscoelastic drops with $Oh_0 = 0.16$ and $Oh_0 = 1.238$, respectively. The deviations of the peak frequency for the first oscillation period from the linear oscillation frequency are 2.8 and 1.7% for the viscoelastic drops with $Oh_0 = 0.16$ and $Oh_0 = 1.238$, respectively. Surprisingly, the peak frequencies are not significantly affected by the change of the viscosity and the two viscoelastic timescales. They rather impact the decay rate, as seen in the previous paragraph. Even the higher frequency corresponding to the mode $k = 4$ is around 7.8 in the diagram for the first oscillation and is even less represented in the power spectrum.

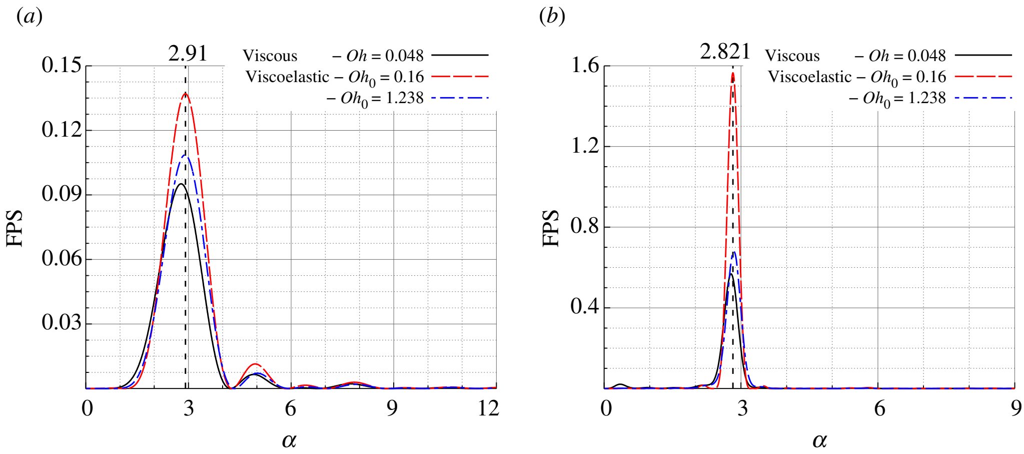

Figure 9. Squared real parts of the Fourier power spectra of the oscillation frequency for the drop north pole motion corresponding to figure 5 for $m = 2$ and $\eta_0 = 0.35$: results in (*a*) for the first oscillation and (*b*) for the interval $0 \leq t \leq 12$. The results include solutions up to the second order.

The spectra evaluated for the time interval $0 \leq t \leq 12$ exhibit secondary peaks below the dominant peak frequency. These frequencies may be interpreted as beat frequencies, which develop with ongoing time and are therefore not visible in the first oscillation period. The frequency of 1.6 appears in all three spectra, and the value around 0.4 is found in the spectra for the two viscoelastic cases. The oscillations from the first period interact with those developing later in time.

(e) State of onset of aperiodic motion

The state of onset of aperiodic behaviour of viscoelastic drops is studied. For Newtonian liquids, in the mode of initial drop deformation $m = 2$, the cut-off Ohnesorge number, where the oscillation frequency vanishes, $Oh_{co,N} = 0.76647$ (notwithstanding the result by Prosperetti [8] for the initial-value problem). Figure 2*b,d* shows that viscoelastic drops may oscillate at far higher Oh_0 than the cut-off value of $Oh_{co,N}$ for Newtonian drops. Viscoelastic drops with, e.g. $De_1 = 33.93$ and the ratio $De_2/De_1 = 0.03714$ of the two Deborah numbers, oscillate at Ohnesorge numbers up to $Oh_0 = 27$. In the present section, first, the Ohnesorge number Oh_0 of viscoelastic drops with varying De_1 and De_2 at the cut-off state is studied. The data are obtained from solutions of the characteristic equation (3.11) of the drop. Figure 10*a* depicts the critical Ohnesorge number $Oh_{0,crit}$ of the viscoelastic drop as a function of the Deborah number De_1 for the value of $De_2/De_1 = 0.1$ as an example. The data are represented empirically by an equation of the form

$$Oh_{0,crit} = a_0 + a_1 De_1^{a_2}, \tag{4.4}$$

where the values of the a_i depend on the ratio De_2/De_1. This dependency may be represented by empirical functions given in the electronic supplementary materials. The data cover the range $0.037 \leq De_2/De_1 \leq 0.2$. For a given Deborah number De_1, the critical Oh_0 decreases with increasing De_2/De_1. The increasing deformation retardation time hinders the oscillatory motion, therefore requiring a drop with lower Oh_0.

Secondly, the cut-off state is characterized by the corresponding decay rate of the drop surface deformation. The difference between the viscoelastic cut-off decay rate and the Newtonian value $\alpha_{r2,co,N} = 2.72338$ is depicted in figure 10*b* as a function of the

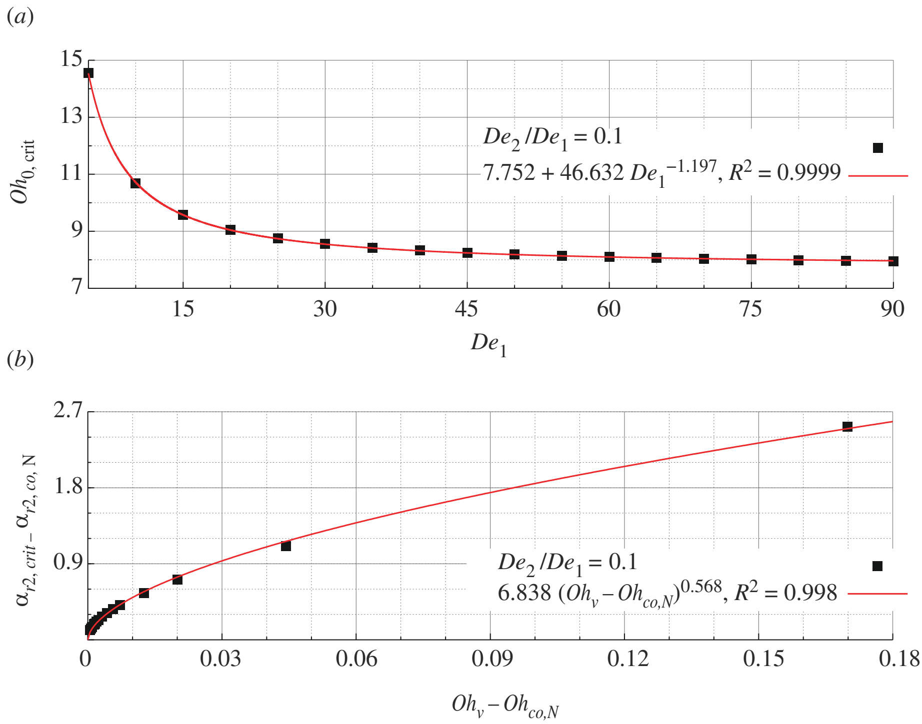

Figure 10. The state of viscoelastic drops at the cut-off of aperiodic motion: (*a*) the critical Ohnesorge number $Oh_{0,\,crit}$ as a function of the drop Deborah number De_1, and (*b*) the decay rate $\alpha_{r2,\,crit}$ of the deformation as a function of the modified drop Ohnesorge number Oh_v, both for the example that $De_2/De_1 = 0.1$ and the mode of initial drop deformation $m = 2$.

difference between the modified Ohnesorge number Oh_v and the Newtonian value at cut-off $Oh_{co,\,N} = 0.76647$ for the example $De_2/De_1 = 0.1$. The values apply to the mode of initial deformation $m = 2$. These critical states to aperiodic drop behaviour are represented empirically by a power law, reading

$$\alpha_{r2,\,crit} - \alpha_{r2,\,co,\,N} = b_1\big(Oh_v - Oh_{co,\,N}\big)^{b_2}. \tag{4.5}$$

The decay rate increases with the modified Ohnesorge number Oh_v. The latter quantity, on the other hand, contains the decay rate, so that the depicted relation is implicit. The coefficients b_1 and b_2 again depend on the ratio of the Deborah numbers De_2/De_1. The dependencies are represented by empirical functions given in the electronic supplementary materials.

With these findings, the oscillatory or aperiodic behaviour of a drop with a given size from a rheologically characterized viscoelastic liquid, behaving according to the Oldroyd-B model, may be predicted. Furthermore, the rate of aperiodic decay of drop surface deformations in the cut-off state is obtained. Knowing all the material data $\mu_0, \sigma, \rho, \lambda_1$ and λ_2 of the drop liquid, the viscoelastic timescale (or Deborah number) ratio, $\lambda_2/\lambda_1 \equiv De_2/De_1$, is known. Using this ratio, the three coefficients a_i in the empirical equation (4.4) for the critical Ohnesorge number may be determined, so that the relationship between the critical Ohnesorge number $Oh_{0,\,crit}$ and the Deborah number De_1 is known. This relationship allows the size d_{crit} to be determined, at which a drop from the given liquid assumes the critical state. Comparing this critical size with the size d of a given drop, allows the drop behaviour to be predicted: drops with $d > d_{crit}$ will perform damped shape oscillations, while deformations of drops with $d \leq d_{crit}$ will decay aperiodically. The critical drop size now allows all the characteristic numbers Oh_0, De_1 and De_2 to be calculated. This allows the decay rate of the drop with the critical size to be determined from equation

(4.5). Knowing all the characteristic non-dimensional numbers governing the shape oscillations, the actual oscillation frequency and decay rate of dampened shape oscillations for a drop with oscillatory behaviour (i.e. with a size $d > d_{crit}$), or the rate of aperiodic decay for a drop with $d \leq d_{crit}$, are obtained as a solution of the characteristic equation (3.11).

5. Conclusions

A weakly nonlinear analysis of axisymmetric shape oscillations of viscoelastic Oldroyd-B drops in a dynamically inert environment was performed. The aim was to reveal the detailed time behaviour of the nonlinear shape oscillations for the two-lobed mode $m = 2$ of initial drop deformation, carrying the derivations to the second order of approximation. The characteristic equation yields an infinite number of solutions for the complex angular oscillation frequency. The appropriate pair of complex conjugate solutions is selected according to experimental data for damped linear shape oscillations of aqueous poly(acrylamide) solution drops in an acoustic levitator. The linear solutions from the theory agree well with the experimental data. The theory reveals two nonlinear effects, which are a time asymmetry and a frequency change for viscoelastic drops with increasing initial drop deformation. The excess time the drop spends in the prolate mode is reduced by more than one-half for the whole range of initial drop deformations in comparison with a viscous drop with the same first-order decay rate. Comparing the trace of the drop north pole motion in time against the corresponding viscous case, elastic effects are found to reduce the minimum drop aspect ratio in the oblate state by around 12% against the viscous drop. The oscillation frequency shows the same trend to vary with increasing initial drop deformation as known from both the inviscid and the viscous cases. The present second-order theory predicts a slight increase in the frequency with the initial drop deformation. A third-order approximation might add to this finding. The variation of the oscillation frequency with the drop deformation makes it vary in time during dampened oscillations, rendering the oscillations quasi-periodic. Likewise, the damping rate of the oscillations varies with the drop deformation, converging to the linear value within the first 1.5 oscillation periods. The Fourier power spectra of the north pole motion show an increase in the peak frequency within the first oscillation period against the solution of the characteristic equation. Secondary peaks appear later in time owing to mode coupling and the consequent beat between the different oscillation mode frequencies. The state of onset of aperiodic motion is characterized by solutions of the characteristic equation, using empirical equations for the cut-off Ohnesorge number $Oh_{0,crit}(De_1, De_2)$ and the decay rate $\alpha_{r2,crit}$. These findings allow the oscillatory or aperiodic motion of a drop with a given size from a rheologically characterized Oldroyd-B liquid to be predicted.

Data accessibility. This article has no additional data.

Supplementary material is available online [47].

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. D.Z.: data curation, investigation, methodology, writing—original draft; G.B.: conceptualization, funding acquisition, supervision, writing—review and editing.

Both authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest. We declare we have no competing interests.

Funding. The authors acknowledge Graz University of Technology for funding the open-access publication of this article.

Acknowledgements. Financial support of this research project by the Austrian Science Fund (FWF) through project number I3326-N32 in the DACH framework is gratefully acknowledged. The authors acknowledge the excellent cooperation with Prof. Dr. M. Oberlack and his group at Darmstadt University of Technology in Darmstadt, Germany.

References

1. Rayleigh JWS. 1879 On the capillary phenomena of jets. *Proc. R. Soc. Lond.* **29**, 71–97. (doi:10.1098/rspl.1879.0015)
2. Lamb H. 1881 On the oscillations of a viscous spheroid. *Proc. Lond. Math. Soc.* **13**, 51–66. (doi:10.1112/plms/s1-13.1.51)
3. Lamb H. 1932 *Hydrodynamics*, 6th edn. Cambridge, UK: Cambridge University Press.
4. Chandrasekhar S. 1959 The oscillations of a viscous liquid globe. *Proc. Lond. Math. Soc.* **9**, 141–149. (doi:10.1112/plms/s3-9.1.141)
5. Reid WH. 1960 The oscillations of a viscous liquid drop. *Quart. Appl. Math.* **18**, 86–89. (doi:10.1090/qam/114449)
6. Miller CA, Scriven LE. 1968 The oscillations of a fluid drop immersed in another fluid. *J. Fluid Mech.* **32**, 417–435. (doi:10.1017/S0022112068000832)
7. Basaran OA, Scott TC, Byers CH. 1989 Drop oscillations in liquid-liquid systems. *AIChE J.* **35**, 1263–1270. (doi:10.1002/aic.690350805)
8. Prosperetti A. 1980 Free oscillations of drops and bubbles: the initial-value problem. *J. Fluid Mech.* **100**, 333–347. (doi:10.1017/S0022112080001188)
9. Khismatullin DB, Nadim A. 2001 Shape oscillations of a viscoelastic drop. *Phys. Rev. E* **63**, 061508. (doi:10.1103/PhysRevE.63.061508)
10. Chrispell JC, Cortez R, Khismatullin DB, Fauci LJ. 2011 Shape oscillations of a droplet in an Oldroyd-B fluid. *Physica D. Nonlinear Phenomena.* **240**, 1593–1601. (doi:10.1016/j.physd.2011.03.004)
11. Hoath SD *et al.* 2015 Oscillations of aqueous PEDOT:PSS fluid droplets and the properties of complex fluids in drop-on-demand Inkjet printing. *J. Nonnewton. Fluid Mech.* **223**, 28–36. (doi:10.1016/j.jnnfm.2015.05.006)
12. Li F, Yin XY, Yin XZ. 2019 Small-amplitude shape oscillation and linear instability of an electrically charged viscoelastic liquid droplet. *J. Nonnewton. Fluid Mech.* **264**, 85–97. (doi:10.1016/j.jnnfm.2018.10.001)
13. Trinh EH, Marston PL, Robey JL. 1988 Acoustic measurement of the surface tension of levitated drops. *J. Colloid Interface Sci.* **124**, 95–103. (doi:10.1016/0021-9797(88)90329-3)
14. Hiller WJ, Kowalewski TA. 1989 Surface tension measurements by the oscillating droplet method. *PCH. Physico. Chem. Hyd.* **11**, 103–112.
15. Stückrad B, Hiller WJ, Kowalewski TA. 1993 Measurement of dynamic surface tension by the oscillating drop method. *Exp. Fluids.* **15**, 332–340. (doi:10.1007/BF00223411)
16. Hsu CJ, Apfel RE. 1985 A technique for measuring interfacial tension by quadrupole oscillation of drops. *J. Colloid Interface Sci.* **107**, 467–476. (doi:10.1016/0021-9797(85)90199-7)
17. Egry I, Lohöfer G, Seyhan I, Schneider S, Feuerbacher B. 1998 Viscosity of eutectic Pd78Cu6Si16 measured by the oscillating drop technique in microgravity. *Appl. Phys. Lett.* **73**, 462–463. (doi:10.1063/1.121900)
18. Perez M, Salvo L, Suéry M, Bréchet Y, Papoular M. 2000 Contactless viscosity measurement by oscillations of gas-levitated drops. *Phys. Rev. E* **61**, 2669–2675. (doi:10.1103/PhysRevE.61.2669)
19. Matsumoto T, Nakano T, Fujii H, Kamai M, Nogi K. 2002 Precise measurement of liquid viscosity and surface tension with an improved oscillating drop method. *Phys. Rev. E* **65**, 031201. (doi:10.1103/PhysRevE.65.031201)
20. Gottier GN, Amundson NR, Flumerfelt RW. 1986 Transient dilation of bubbles and drops: theoretical basis for dynamic interfacial measurements. *J. Colloid Interface Sci.* **114**, 106–130. (doi:10.1016/0021-9797(86)90244-4)
21. Tian Y, Holt RG, Apfel RE. 1995 Investigations of liquid surface rheology of surfactant solutions by droplet shape oscillations: theory. *Phys. Fluids (1994).* **7**, 2938–2949. (doi:10.1063/1.868671)
22. Apfel RE *et al.* 1997 Free oscillations and surfactant studies of superdeformed drops in microgravity. *Phys. Rev. Lett.* **78**, 1912–1915. (doi:10.1103/PhysRevLett.78.1912)
23. Tian YR, Holt RG, Apfel RE. 1997 Investigation of liquid surface rheology of surfactant solutions by droplet shape oscillations: experiments. *J. Colloid Interface Sci.* **187**, 1–10. (doi:10.1006/jcis.1996.4698)

24. Aske N, Orr R, Sjöblom J. 2002 Dilatational elasticity moduli of water-crude oil interfaces using the oscillating pendant drop. *J. Dispers. Sci. Technol.* **23**, 809–825. (doi:10.1081/DIS-120015978)
25. Lu HL, Apfel RE. 1991 Shape oscillations of drops in the presence of surfactants. *J. Fluid Mech.* **222**, 351. (doi:10.1017/S0022112091001131)
26. Zhang XG, Harris MT, Basaran OA. 1994 Measurement of dynamic surface tension by a growing drop technique. *J. Colloid Interface Sci.* **168**, 47–60. (doi:10.1006/jcis.1994.1392)
27. Kovalchuk VI, Krägel J, Aksenenko EV, Loglio G, Liggieri L. 2001 Oscillating bubble and drop techniques. In *Novel methods to study interfacial layers*, pp. 485–516. Amsterdam, Netherlands: Elsevier.
28. Ravera F, Loglio G, Kovalchuk VI. 2010 Interfacial dilational rheology by oscillating bubble/drop methods. *Curr. Opin. Colloid Interface Sci.* **15**, 217–228. (doi:10.1016/j.cocis.2010.04.001)
29. Yang L *et al.* 2014 Determination of dynamic surface tension and viscosity of non-Newtonian fluids from drop oscillations. *Phys. Fluids (1994).* **26**, 113103. (doi:10.1063/1.4901823)
30. Brenn G, Plohl G. 2015 The oscillating drop method for measuring the deformation retardation time of viscoelastic liquids. *J. Nonnewton. Fluid Mech.* **223**, 88–97. (doi:10.1016/j.jnnfm.2015.05.011)
31. Brenn G, Teichtmeister S. 2013 Linear shape oscillations and polymeric time scales of viscoelastic drops. *J. Fluid Mech.* **733**, 504–527. (doi:10.1017/jfm.2013.452)
32. Ponce-Torres A, Montanero JM, Herrada MA, Vega EJ, Vega JM. 2017 Influence of the surface viscosity on the breakup of a surfactant-laden drop. *Phys. Rev. Lett.* **118**, 024501. (doi:10.1103/PhysRevLett.118.024501)
33. Hosseinzadeh VA, Brugnara C, Holt RG. 2018 Shape oscillations of single blood drops: applications to human blood and sickle cell disease. *Sci. Rep.* **8**, 16794. (doi:10.1038/s41598-018-34600-7)
34. Tamim SI, Bostwick JB. 2021 Oscillations of a soft viscoelastic drop. *NPJ. Microgravity.* **7**, 42. (doi:10.1038/s41526-021-00169-1)
35. Zrnić D, Brenn G. 2021 Weakly nonlinear shape oscillations of inviscid drops. *J. Fluid Mech.* **923**, A9. (doi:10.1017/jfm.2021.568)
36. Zrnić D, Berglez P, Brenn G. 2022 Weakly nonlinear shape oscillations of a Newtonian drop. *Phys. Fluids (1994).* **34**, 043103. (doi:10.1063/5.0085070)
37. Calvert P. 2001 Inkjet printing for materials and devices. *Chem. Mater*. **13**, 3299–3305. (doi:10.1021/cm0101632)
38. Pandey K, Prabhakaran D, Basu S. 2019 Review of transport processes and particle self-assembly in acoustically levitated nanofluid droplets. *Phys. Fluids (1994).* **31**, 112102. (doi:10.1063/1.5125059)
39. Kim J. 2007 Spray cooling heat transfer: the state of the art. *Int. J. Heat Fluid Flow.* **28**, 753–767. (doi:10.1016/j.ijheatfluidflow.2006.09.003)
40. Breitenbach J, Roisman IV, Tropea C. 2018 From drop impact physics to spray cooling models: a critical review. *Exp. Fluids* **59**, 55. (doi:10.1007/s00348-018-2514-3)
41. Kamperman T, Trikalitis VD, Karperien M, Visser CW, Leijten J. 2018 Ultrahigh-throughput production of monodisperse and multifunctional janus microparticles using in-air microfluidics. *ACS Appl. Mater. Interfaces* **10**, 23433–23438. (doi:10.1021/acsami.8b05227)
42. Baumgartner D, Brenn G, Planchette C. 2020 Effects of viscosity on liquid structures produced by in-air microfluidics. *Phys. Rev. Fluids* **5**, 103602. (doi:10.1103/PhysRevFluids.5.103602)
43. Smuda M, Kummer F, Oberlack M, Zrnić D, Brenn G. 2024 From weakly to strongly nonlinear viscous drop shape oscillations: an analytical and numerical study. *Phys. Rev. Fluids.* **9**, 1–30. (doi:10.1103/PhysRevFluids.9.063601)
44. Bird RB, Stewart WE, Lightfoot EN. 1962 *Transport phenomena*. New York, NY: John Wiley & Sons.
45. Tomotika S. 1935 On the instability of a cylindrical thread of a viscous liquid surrounded by another viscous fluid. *Proc. R. Soc. Lond. A.* **150**, 322–337. (doi:10.1098/rspa.1935.0104)
46. Renoult MC, Brenn G, Plohl G, Mutabazi I. 2018 Weakly nonlinear instability of a Newtonian liquid jet. *J. Fluid Mech.* **856**, 169–201. (doi:10.1017/jfm.2018.677)

47. Zrnić D, Brenn G. 2024 Data from: weakly nonlinear shape oscillations of viscoelastic drops. Figshare. (doi:10.6084/m9.figshare.c.7399609)